



THE HANDBOOK FOR ETRM

Start Your Learning In Energy Trading
And Risk Management



MITTAL SHAH

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And Risk Management

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INDIA • SINGAPORE • MALAYSIA



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EXW – Ex-Works or Ex-Warehouse

FCA – Free Carrier

FAS – Free Alongside Ship

FOB – Free On Board

CFR – Cost and Freight

CIF – Cost, Insurance and Freight

CPT – Carriage Paid To

CIP – Carriage and Insurance Paid To

DAP – Delivered At Place

DPU – Delivered At Place Unloaded

DDP – Delivered Duty Paid

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FOREWORD

Be it the Ukraine war, the pandemic when oil Futures prices in the US turned negative, the oil price crash after the global financial crisis, the gulf crisis or the oil embargo: The focus is on Oil. It impacts all lives as it is an important energy source. The producer and the consumers reside in different areas. Thus, making oil one of the most traded commodities. Moreover, most of the oil trading happens in an Over-The-Counter (OTC) market.

Though it is necessary, the knowledge of oil/energy trading is minimal. For example, people wonder, with future prices negative, why petrol pumps are not giving oil for free. Why does the US publish their oil reserve data regularly?

Throughout my academic and professional career, I needed help finding a reference book as an essential go-to book to understand the concept. A student graduating with a focus on capital markets does not have an in-depth understanding, nor does an IT analyst working in the area have an overall understanding. The lack of knowledge leads to a demand for talent. That increases the earning potential of professionals as all professionals are trading their hours for money.

Since no single book explains OTC energy trading to a student, the only way is to go through multiple expensive books for one topic or training to develop expertise in energy trading. The other approach is time-consuming on-the-job learning.

Hence, I decided to write a book based on my experience and challenging journey in learning energy trading. The book aims to help

students studying with a significant focus on Energy and IT professionals work on clients involved in commodities trading, which can explain almost all concepts with easy-to-understand language.

The book takes a trade life cycle as a base to explain concepts of energy trading. Other products like electricity, natural gas, refined products, and coal are also traded under energy trading. The commodities' trade life cycle is handled in a separate chapter. The book also has a section on current trends in technology in the field.

My humble request is that if you want to learn energy trading as a retail investor, this is not the book for you, so if you have purchased it, please return it.

Let me take this opportunity to thank you for purchasing the book and deciding to read it. I wish you all the best for your great professional and academic future and happy learning.

HOW TO READ THE BOOK

This is neither fiction nor a self-help book. Neither is it a management book. The book fulfils a gap I identified in my career. The gap identified was a single book that gives an idea of energy trading and risk management. The book will help students and IT professionals understand the basics. The book can help evangelist who wants to understand proprietary trading.

A reader with a basic understanding of capital markets and supply chains can draw immense value from the book. I have tried to include as many basics as possible. Some of the concepts of capital markets and supply chains are covered in the chapters—an additional chapter: “Back to Basics” for more fundamentals. The last chapter: “Jargon and Definition”, cover other basics that could not be covered within the chapters due to continuity. Still, the reader mostly likes to hear about the energy trading and risk management world.

All topics covered are vast. One can specialize in each case. Moreover, the book is based on experience and on-the-job learning. Most of the sources of information are from the internet and some books. So, to learn more about each topic, I suggest searching the internet. I want to take this opportunity to thank my business users, client managers, managers, supervisors and team members for sharing their knowledge and experience to enrich my knowledge.

Lastly, the book has some stories which will keep the readers interested.

INTRODUCTION TO ENERGY TRADING

1.1 History of the Modern Oil Industry

The modern oil industry started with the discovery of oil in Baku in 1837. The first modern oil well reached a depth of 21m in 1846. It would come as a surprise to everyone that the middle east was not where the industry expanded. Oil was discovered in Kuwait and Saudi in 1938.

It was Europe, Canada and the US where the commercial production of oil started. It started with the oil wells in Bobrka, Poland, in 1854. Ignacy Łukasiewicz founded the world's first oil mine in Bobrka. Bucharest, Romania followed it in 1857. Commercial oil production moved to the American continent in 1858, with production in Canada and Pennsylvania.

The first successful use of a drilling rig to extract oil was in 1859 by George Bissell and Edwin L Drake. In 1901, oil was discovered in Beaumont, Texas, by Captain Anthony Francis. The first thing to flow out of the well, which was at a depth of 370m, was natural gas. The oil that flowed out spurted to a height of 60m and lasted nine days.

The first oil company that controlled 90% of the oil market by 1890 was Stat Oil, incorporated by John D. Rockefeller. In 1911, the first effort to break the monopoly of big oil companies was not long after that when Stat Oil was split into 33 smaller companies.

As mentioned above, oil was discovered in Kuwait and Saudi Arabia in 1938. The Suez crisis threatened supply to Europe in 1956.

Understanding the importance of controlling production to influence prices OPEC (Organization of the Petroleum Exporting Countries) was formed in 1960. The Arab Oil Embargo brought to light the importance of securing oil production and supply in 1973, and no wonder oil is one of the most actively traded commodities.

The emergence of oil as a trading asset class gained traction with starting of Futures trading on the New York Mercantile Exchange (NYMEX) in 1978.

Wars in the middle east from 1980 to 1990 kept oil prices volatile. 1997 exposed the correlation between the financial crisis impacting oil prices.

In 2008, just before the global financial crisis, oil reached the price of \$147.27 per barrel and crashed approximately 78% of its peak price. December 2020 was the first year when the US became a net oil exporter since 1949, adding new dynamics to oil prices and trading. The COVID pandemic and war in Ukraine again took oil and gas trading to new horizons and reiterated the importance of the commodity.

Such is the importance of oil and now energy that it is the highest trading commodity in the world. In the subsequent section, we will understand the advantage of energy trading and detail energy trading done by traders (not retail).

1.2 Trading

Trade can be defined as “A voluntary exchange of goods and services between economic actors such as consumers, companies and nations.”

Trading is an activity done by producers and suppliers. Commodity trading is a robust and fast-growing industry. Oil is the largest traded commodity in the world. Energy trading offers investors the following advantages. Most of the advantages apply to other commodities also and hence the examples include different commodities.

1.3 Advantages of Commodities/Energy Trading

Following are some of the most relevant advantages of Commodities/Energy Trading

- (a) **Potential hedge against inflation:** When an investor looks to invest in something, they always consider the returns if the returns beat inflation to make profits. During inflation, products like pulses, oil, gold, cotton etc., get expensive. So, when one is investing in these commodities by engaging in Futures trading, they beat inflation. Gold is a valuable hedge against inflation.
- (b) **Diversification of your portfolio:** For an investment portfolio, diversification is risk management. An asset portfolio has many different kinds of assets. Some of the asset classes are equity, debt, real estate and commodities. Commodities are an independent asset class because commodity prices are vulnerable to market forces like supply-demand, weather conditions, geopolitical tensions, foreign exchange rates and natural disasters.
- (c) **High leverage:** Investors can take a position by buying underlying commodities. For example, gold can be purchased and stored. The asset is used as collateral in need of liquidity. But a significant amount of commodities trading happens in the

Futures market. One can take a position by paying only a fraction of that value as a margin. Generally, it has been observed that the margin requirement in the commodity Futures market is lower than in equity Futures and Options.

- (d) **Less manipulation:** Prices are determined by global markets as commodities are traded globally. Moreover, it is difficult to corner the supply/demand of a particular commodity. The local prices can be influenced in a limited way by local factors global as supply and demand dynamics take over. Since the commodities market is universal, the prices are less susceptible to manipulation.

If a company or a country is an importer and exporter, it provides the following benefits:

Hedging of price risk: Commodities' prices are highly volatile. Since commodities are an independent asset class, their price movements are also different from other asset classes. Moreover, commodity prices are susceptible to supply chain disruptions. Business, while buying or selling, wants to know the cost of production. Due to high volatility, it becomes difficult to determine the cost of production. Example: For an airline, one of the significant costs of operation is aviation fuel. Hence an airline is vulnerable to sudden price changes in oil. The airline would want to reduce its exposure to fluctuating oil prices. Thus airlines are exposed to price risk. Similar to the airline industry, many industries are exposed to physical commodities; all of them carry a price risk. To reduce this price risk, they can get involved in Futures trading or get into an Over-The-Counter agreement that

allows them to procure or sell commodities at a price based on a formula, thereby reducing the impact of sudden price movements on their businesses. Commodities trading helps the industry manage price risk effectively.

The following section looks at the needs that drive a producer and consumer of oil or other commodities approach to trading.

A **producer** of the Futures can help you in the following ways:

Lock-in-price for your product: Let us understand by taking the example of oil producers. For an oil producer, the cost of operation is nearly fixed once oil is discovered and the oil rig is set up onshore or offshore. The price of oil keeps changing in the spot and Futures markets. Now, if the oil producers wait to sell the product in real-time, first, it is logistically not possible and second, there are chances the producer makes a loss. Now, if the oil producers sell the oil in future markets, the price is fixed. Then to increase the profitability the oil producer needs to manage the cost of the operation.

Another example to understand is the farmer for whom the cost of a particular crop, fertilizer, labour, seeds, water and energy are generally fixed. But the prices of farm produce keeps fluctuating. If the farmer waits till harvest and lets the market determine the price, then the farmer is exposed to price risk. The farmer can sell the future of grains and can lock in the price of produce. Similarly, other industries, for example, gold, metals mining companies or oil producers, can lock in their rates by taking a position in the commodities trading market.

Guaranteed demand: Continuing the above example, an oil producer would need to deliver the quantity of oil as per the Futures contract and hence continuous demand. If production is high, then the oil producer only has to find a buyer on the spot for the extra oil produced. Similarly, in the case of the farmer, let us assume the farmer has produced 100 metric tons of grains. The farmer goes to the local market to sell, but the demand is 90 metric tons. The farmer has to wait for a buyer for ten metric tons to get a profitable price or sell it at a lower price. By taking a future position for an estimated production, the farmer can sell all the products without worrying about the demand on the day of harvest. Thus, by selling the contract, the farmer has guaranteed a market for the produce.

A **consumer** of oil or any other commodity, trading helps in the following ways:

Control your cost: Let us take the jeweller as an example. A jeweller makes ornaments and is a consumer of gold. Apart from the gold price, the rest of the costs, like rent, employee cost, and marketing, is stable. Thus, buying knowing the gold price, a jeweller can determine the total cost of production of the end product. A sudden change in the price of gold cannot impact the company's liquidity requirements. Such awareness regarding the cost of production will help jewellers in composition.

Another example is that of an airline. If the price of oil increases, there is limited capacity to pass on the prices to consumers. A sudden surge in oil prices directly impacts the airline industry's profitability.

Thus, a producer would protect the market and margin by using commodities Futures trading.

Ensure uninterrupted supply: Commodities are an asset class that are susceptible to supply chain disruption. As a consumer of the commodity, ensuring the supply of the raw material is paramount. One of the advantages of Futures trading in a commodity market is that exchange will ensure the delivery of the commodity. Taking a position reduces supply chain disruption risk. An uninterrupted supply can be determined by buying Futures in the commodity markets. By doing the same, the producer will assure the supply of your raw materials for a fixed quantity at a pre-determined price at the scheduled time. For example, an oil refinery will always need oil. To ensure uninterrupted supply, the oil refinery owner should buy oil Futures.

From the above example of a farmer, oil producer, gold jewellery maker or oil refinery we can say with confidence that commodities trading helps both the producer and consumer of commodities.

Energy is one of the basic needs of human beings. Oil trading takes on multiple dimensions. These dimensions are different from a financial market. By understanding the multi-dimensional approach, one can understand more about oil trading.

1.4 Oil Trading: A multi-dimensional approach

The different types of asset classes available for investment or trading are (a) Cash, (b) Equity, (c) Fixed Income, (d) Real Estate and (e) Commodities. Before we progress and criticize this asset classification, there are multiple ways to do asset classifications. One such example

of other asset classification is (a) Cash and Cash Equivalent (b) Equity (c) Fixed Income (d) Real Estate (e) Commodities and (f) Luxury (art, car, wines etc.). Let us take a look at what these asset classes represent.

- (a) **Cash:** The purpose of cash is to provide liquidity and the power to buy. The different forms/examples of cash are liquid funds, saving accounts, online wallets and hard cash.
- (b) **Equity:** Equity represents ownership of a business. The ownership is in the form of stocks (shares). The other related instruments are ownership-based startup funding, index funds, exchange-traded funds and equity mutual funds, to name a few.
- (c) **Fixed Income:** Fixed income as an asset class be understood as lending money to someone, be it individual or corporate. Some products are fixed deposits, recurring deposits, bonds, debentures, debt mutual funds etc.
- (d) **Real Estate:** This asset class represents the ownership of physical space. Some instruments that help in investing in real estate are flats, plots, bungalows, and commercial or REIT (Real Estate Investment Trust).
- (e) **Commodities:** Commodities as an asset class represent an asset class related to the ownership of goods with an end-user. Some examples of commodities in the asset class are gold, silver, copper, crude oil, wheat, and corn, to name a few.

Of all the above asset classes, the dimensions involved in trading commodities are the maximum. Oil is one of the commodities that is consumed and traded extensively. The dimensions involved in oil

trading can be macro or micro. Following are some of the dimensions of oil trading.

- (a) **Geopolitical:** Conflicts in areas of production or sanctions on oil-producing countries can significantly impact oil supply and thus impact oil pricing. The other dimension is the strategic oil reserves of countries. For example, the US has 700 million barrels of oil stored in strategic reserves, and sometimes it buys oil to fill the reserves or releases oil from the strategic reserves. This decision can be political. Hence geopolitics plays have an impact on oil pricing.
- (b) **Benchmarks:** Oil is generally traded in the Futures market based on benchmarks. The spot price of the oil is either premium or discounted to these benchmarks. The factors affecting the same can be purely local. During the COVID-19 pandemic, the settlement prices of WTI (Western Texas Intermediate) turned negative as all the local storages were full. Such an impact on a particular benchmark can lead to cascading effects on other benchmarks.
- (c) **Demand volatility:** The demands of the consuming region can change suddenly. For example, cold weather might reduce travel, reduce the need for motorable oil (petrol/gasoline), and increase heating oil. Local developments like disruption at a port or a strike also change demand.
- (d) **Location and Logistics:** Efficient logistics is one of the backbones of this competitive field. The blocking of the Suez Canal in 2021 suddenly disrupted the oil supply from the middle

east to refineries in Europe. Such disruption leads to volatility in prices, which might be temporary but cannot be predicted easily.

- (e) **Product Specification:** Certain oil grades are produced, and refineries consume a different oil quality. Thus the product specification changes between production and consumption. Moreover, blending is employed in the refinery to achieve an accurate product mix to reduce cost. Depending on the blending requirements as the refinery gets configured leads to a change in demand. It is challenging to know the blending process a refinery would adapt to, thus adding uncertainty to demand.
- (f) **Cost of Funds:** The trade finance division of banks funds the oil trade. Different banks depending on the ability or risk profile of their customers offer rates for financings. Thus, the cost of funds determines the transaction's profitability, thus leading to a different number of trades from a particular player in oil trading, thus increasing or reducing the market depth.
- (g) **Futures Market:** Most oil trading happens in the Futures market. Future market pricing helps in predicting the pricing based on the information available. But we know that future trading can happen in Contango or Backwardation. The peculiar behaviour of commodities prices leads to different trading strategies. For example, suppose the market is in Contango, and the storage cost is less. In that case, participants might want to buy and store the crude, thus impacting supply in the current and future demand, leading to an additional dimension in pricing.

- (h) **Counterparty and Political Risk:** Oil trade transactions are large-scale. Thus, active monitoring of the counterparty risk becomes paramount. The sovereign risk associated with the counterparty adds another element of risk. Pricing in such risk also impacts oil pricing in spot and Futures markets as different players have different risk appetites.
- (i) **Alternate sources and substitutes:** Different production sites have different cost structures. Extracting oil from the middle east is different from oil sands in Canada. Thus, prices determine the profitability of production sites. Some of the production sites can be brought onstream at very short notice leading to uncertainty on supply augmentation directly impacting the prices.
- (j) **Cost/Availability of Freight:** A fixed number of tankers worldwide can transport oil. At the same time, the storage space is also fixed. Efficient logistics, as mentioned, is paramount in making a profitable transaction. A sudden glut or supply of tankers changes the cost of transport, which impacts the price of overall trade as freight cost directly moves the profitability of the trade.

The ten dimensions directly impact supply and demand. Most of the dimensions are unpredictable. Hence not oil, commodities as an asset class is highly volatile. At the same time, prices can be different in different markets. None of the other asset classes has so many dimensions to look at while booking a trade. These dimensions make oil trading a multi-dimension discipline. Lastly, these dimensions are

relevant for other commodities traded, but their importance, relevance and impact would differ.

The multi-dimensional feature of energy trading attracts different types of players in the market. The different types of players are explained in detail in the next section.

1.5 Type of players in energy trading

The players in Energy Trading can be classified as (a) Hedgers, (b) Speculators and (c) Arbitrageurs.

Hedgers are traders or players who can be individuals or organisations who are users of energy. The changes in prices impact the users of energy. The changes in prices affect the bottom line. Hence, they come to the market to fix the price or reduce the impact of price changes. They are primary participants in the Futures or Forward markets. For example, a producer of Oil knows the cost of production of oil. The changes in the prices in spot markets have a direct impact on profitability. To reduce the effect of the volatility in prices, they sell Futures contracts or Forward contracts to hedge against price fluctuations. Many types of hedgers come to the market. The primary types of hedgers are buy-side hedgers and sell-side hedgers. Buy-side hedgers are consumers (refineries) worried about a price increase. The sell-side hedgers are generally producers (oil fields and upstream), and they are concerned about the decrease in prices.

Speculators are individuals or firms that are interested in making profits from price fluctuations. The risk appetite of speculators is high. They are neither producers nor consumers. They are interested in price fluctuations. Individual speculators generally operate in the

Futures market and primarily trade on exchanges. The proprietary trading firms are participants in both exchange and OTC markets. Speculators bring liquidity to the market. Liquidity is an essential parameter for a market. Speculators take the risk that the hedgers want to offload.

Arbitrageurs are players that try to make a risk-free profit out of the pricing inaccuracies in the market. For example, the price of Brent crude is \$60 a barrel in Dubai, UAE. The price of Brent crude is \$70 in Mumbai, India. The cost of finance, transportation, insurance, inspection etc., is \$5 per barrel. Then an arbitrageur will buy oil in Dubai and sell it in Mumbai. The example is crude but shows that trade is risk-free for the player, and due to price inaccuracies in the market, they make a profit of \$5 per barrel. The arbitrageurs make the market efficient by keeping the prices honest. They are important players as they try to find inaccuracies in the market and trade. There are different types of arbitrage opportunities available in the market, such as Geography (geographic arbitrage identifies temporary price anomalies between different locations), Time (time arbitrage seeks to benefit from the shape of the Forward curve for physical delivery. For example, in Contango markets, when investors pay a premium for Forwarding delivery, arbitrageurs try to cash and carry arbitrage) or technical (technical arbitrage seeks to benefit from the different pricing perceptions of varying oil grades). Specialised firms with access and presence in different markets can indulge in arbitrage trade.

All three types of players are essential to make an efficient energy

trading market.

As oil trading is multi-dimensional and attracts different players, the pricing behaviour of the markets is different. The uniqueness of commodity/energy trading is explained in the following section.

1.6 Uniqueness to energy trading: Contango and Backwardation

Two terms are often used in energy trading to describe the prices. The two terms are (a) Contango and (b) Backwardation. There is always a difference between the spot price and Futures prices. The difference arrives due to the cost of carrying, assumption of supply of underlying at delivery time, and interest rates, to name a few.

The phenomenon described as Contango happens when the price of a Futures contract is higher than the price of the underlying in the spot market. For example, the spot price of crude is \$80 per barrel, but the future contract price to be delivered after one month is \$90 per barrel. The market in this scenario is called to be in Contango. The characteristics of the market in Contango are (a) a sustained increase in prices and (b) investors have faith that the uptrend will continue in the long term. If crude oil is Contango, it may indicate an immediately available supply.

Similarly, Backwardation happens when the future contract price is lower than the underlying price in the spot market. For example, the oil price for future contracts to be delivered after one month is \$100 per barrel. The price of oil in the spot market is \$90 per barrel. The market, in this case, is called to be in Backwardation, which may happen when the underlying asset is in short supply in the spot

market, but there is an expectation of increased supply in future. Anything threatening the steady flow of oil worldwide, such as imminent war, tends to drive the oil market into Backwardation.

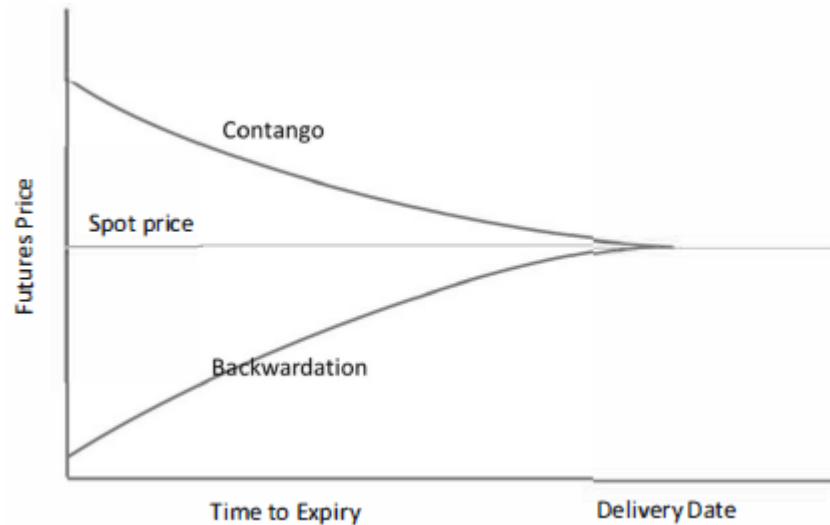


Figure: Convergence of Contango and Backwardation at Expiry

Future contract prices will converge towards the spot price and should converge at the spot price on the delivery date. If such a scenario does not happen, there are opportunities for arbitrage in the underlying market.

Chapter Summary

- Trading can be summarized in the following diagram:

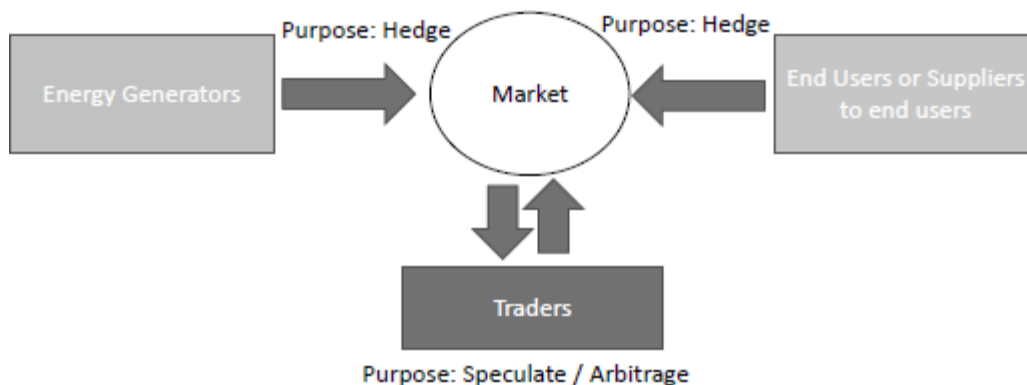


Figure: Trading in a diagram

- Oil trading has advantages as it is a potential hedge against inflation, diversification of your portfolio, high leverage, and less manipulation.
- Oil trading helps to lock-in-price for your product, ensure guaranteed demand, control cost and ensure uninterrupted supply.
- Oil trading is multi-dimensional. The dimensions are Geopolitical, Benchmarks, Demand volatility, Location and Logistics, Product Specification, Cost of Funds, Futures Market, Counterparty & Political Risk, Alternate sources & substitutes and Cost/Availability of Freight.
- Players in oil trading can be classified into three categories. The three categories are (a) Hedgers, (b) Speculators and (c) Arbitrageurs.
- Oil trading has unique terms to describe the pricing. The two terms are (a) Contango and (b) Backwardation.

ENERGY TRADING MARKETS AND PRODUCTS

2.1 Energy Trading Markets

There are two types of trading markets. The first is **(a) Primary Market and (b) Secondary Market**. A primary market is a market where security is created/new and sold to the investors. For example, a company's "Initial Public Offering (IPO)" is a primary market operation. A mutual fund's "New Fund Offer (NFO)" is also a primary market operation. A company issuing debt to investors is also a primary market operation. The secondary market is where these securities are traded after the primary market operation is complete. The secondary market can be broken down into two additional sub-categories (a) Auction and (b) Dealer Markets. An auction is a place in the open outcry system where buyers and sellers meet. A stock exchange like New York Stock Exchange (NYSE) is an example of an auction place. In dealer markets, though, people trade through electronic networks. The oil trading market falls in the secondary market category.

The other classification of an oil trading market is defined as **(a) Exchange Traded and (b) Over the Counter (OTC)**. An exchange is a formally established place where listed securities are traded with defined market participants. The securities are defined, and the exchanges develop rules. Some of the leading exchanges for energy are Intercontinental Exchange (ICE), European Energy Exchange (EEX),

Chicago Mercantile Exchange (CME), New York Mercantile Exchange (NYMEX), Dubai Mercantile Exchange (DME), Multi Commodity Exchange (MCX), Tokyo Commodity Exchange (TOCOM).

An Over-The-Counter is a decentralized market where participants trade directly without a central agency like an exchange or a broker. Generally, there are dealers in the OTC market. As defined, exchanges are centralized places which mean they are structured. OTC trades are contracts between two counterparties directly and hence are entirely unstructured.

The significant difference between trading on an Exchange or OTC can be summarized as follows:

Parameters	Exchange	Over-The-Counter (OTC)
Customization	None. The contracts/securities traded are defined by the Exchange.	The contracts traded are 100% customizable to the needs of the parties.
Credit Risk	The Exchange bears the risk involved in the execution of a trade.	The risk is borne by the parties involved in the trade.
Time	Exchanges operate on a fixed schedule, meaning an exchange has a set time to open and close.	24x7. It is dependent on the parties involved in the trade or the dealer making the trade.

Rules and Regulations	The parties involved need to adhere to the rules of the exchange. Moreover, they are governed by the regulations of the geography in which the exchange is located.	The trade is governed by the rules defined in the contract. There are light regulations in which the trade is getting executed.
Price Discovery	The prices for trading are provided by the exchange. The parties involved do not generally know each as trade matching is anonymous.	The price discovery is done based on the negotiation of the parties involved. The parties know each other before they get into a trade.

Table: Comparison between OTC and Exchange

One more way to describe the oil trade market is (a) Spot markets and (b) Futures markets. A spot market is where an oil trade happens for immediate delivery. Generally, it is an Over-The-Counter market transaction. In the Futures market, the trades occur for delivery in the future. Futures markets consist of both exchange and Over-The-Counter. While the spot trade happens with spot prices, the future trades happen with the future price curve of oil. The oil prices in the Futures market can be in Contango or Backwardation.

In summary, there are many ways to describe an oil trade market, as it is a combination of different classifications of the markets as

described above.

2.2 Energy Trading Products

Some of the common products involved in energy trading are as follows:

2.2.1 Spot Trade

The Spot trades are one-off transactions of large physical cargoes or oil parcels for immediate delivery. The spot crude trade forms a tiny portion of all the oil trading but provides the daily prices and helps set prices for other trades. The Spot trades are generally Over-The-Counter (OTC) between two counterparties but are highly standardized. The spot trade contains information like location (the place where the shipment needs to be delivered), delivery terms, mode of transport, and the period by which the oil needs to be delivered, along with the price, quantity, and quality of the oil.

2.2.2 Forward Contract

A Forwards contract is an agreement between two counterparties exchanging a cargo volume at a specified future date for a specified price. A Forward contract is not traded on an exchange but is an Over-The-Counter (OTC) product. A typical oil contract can look like a refinery will buy 100000 barrels of oil every month for a year at \$60 a barrel. A simple example can be a refinery that will buy 100000 barrels of oil for the next month for a year at \$60 a barrel. In summary, Forward contracts are entirely customizable contracts as they are agreed upon by the parties directly. All the parameters of the trades are customizable, primarily the quantity, quality, price, or the

method to arrive at the price and hence the value of the contract, and the date, price and terms of delivery.

2.2.3 Futures Contract

A Futures contract is a standardized agreement to buy or sell oil at a future date. The Futures contracts are standardized contracts for quantity, quality, delivery date, place, and method of delivery as per the delivery terms defined in the contract. Futures contracts are defined by an exchange and traded on an exchange. Typical contract terms for a Futures contract traded on an exchange are as follows:

Terms of Futures contracts	Description of Terms of Futures Contract
Description	Describes the contract—for example, Exchange for physical or cash settlement for Brent crude.
Trading Screen Name	The name of the contract will appear on the trading screen.
Contract Size	The quantity of underlying. For example, 1000 barrels.
Unit of Trading	The units of barrels that can be traded. For example, Multiple of 1000s. That means if a buyer wants to buy 2000 barrels of oil, then there is a need to buy two Futures contracts.
Currency of Trading	USD. For Energy trading, the prices are always in USD.

Trading Price	The price at which the contract will trade. It is specified in terms of the decimal. For example, 0.001 cents per barrel.
Settlement Price	The price at which the Futures contract will be settled. It is specified in terms of the decimal. For example, 0.001 cents per barrel.
Minimum Price Fluctuation	The minimum price fluctuation that is needed to enter a trade. The price fluctuation is specified in terms of the decimal. For example, 0.001 cents per barrel.
Expiration Date	The day on which the contract will be settled. The exchange specifies a typical day. For example, an exchange can specify as the third Wednesday of every month. The exchange will also set what day to consider if the settlement date falls on a holiday.
Contract Security	The contract specifies the financial guarantee a trader needs to provide to exchange to trade the contract. It can be looked at as an Initial margin required to enter the trade.
Daily Settlement Price	It specifies the time or duration of the time during trading hours that would be considered the contract's settlement price. For example, an exchange can set the average price of the last half hour of trading.

Daily Margin	The Mark-To-Market margin that a trader needs to provide to exchange is due to a settlement price change.
Trading Method	It specifies if the contract can be settled as delivery of the underlying oil or cash.
Delivery/Settlement Terms	It specifies the terms of settlement of the contract.
Position Limits	It specifies the maximum number of contracts a trading entity can buy/sell. The position limit is vital as it does not allow a trading entity to corner the market.

Table: Typical contract terms of a Futures contract

Since a Futures contract is traded on an exchange, each exchange would have its terms defined for a contract. The contract terms and their definition can vary as per the exchange.

2.2.4 Options

An Option is a contract that provides the buyer of the contract the right but not the obligation to buy or sell an underlying oil at a specific price at a specified date in the future, depending on the Option type. A buyer of the “Call” Option has the right but not the obligation to buy the underlying at a specific price at a specified date from the seller of the “Call” Option. Similarly, a buyer of the “Put” Option has the right but not obligation to sell the underlying at a specific price at a specified date to the seller of the “Put” Option. Options are generally traded on an exchange. Similar to a Futures contract, the exchange

defines the terms of an Option contract. Typical contract terms for a Futures contract traded on an exchange are as follows:

Terms of Options contracts	Description of Terms of Options Contract
Description	Describes the contract—for example, Exchange for physical or cash settlement for Brent crude.
Trading Screen Name	The name of the contract will appear on the trading screen.
Option Type	Defines the Option type. It can be a “Call” or a “Put”. It also specifies if it is an American or Bermudan, or Asian style—for example, the American call Option for Brent Crude.
Contract Size	The quantity of underlying. For example, 1000 barrels.
Unit of Trading	The units of barrels that can be traded. For example, multiples of 1000s.
Currency of Trading	USD. For energy trading, the prices are always in USD.
Trading Price	The price at which the contract will trade. It is specified in terms of the decimal. This is called a premium that the buyer would pay to the seller for purchasing the contract. For example, 0.001 cents per barrel.

Settlement Price	The price at which the Options contract will be settled. It is specified in terms of the decimal. For example, 0.001 cents per barrel.
Minimum Price Fluctuation	The minimum price fluctuation of the premium that is needed to enter a trade. The price fluctuation is specified in terms of the decimal. For example, 0.001 cents per barrel.
Expiration Date	The day on which the Options contract will be settled. The exchange specifies a typical day. For example, an exchange can be set as the third Wednesday of every month. The exchange will also determine what day to consider if the settlement date falls on a holiday.
Strike Price Increment	The exchange defines the increment in terms of the strike price. As per the definition, there can be multiple strike prices possible. For example, it can be defined as \$1 per barrel.
Daily Settlement Price	It specifies the time or duration of the time during trading hours that would be considered a settlement price for the contract. For example, an exchange can select the average price of the last half hour of trading. The settlement price is used for the daily margin for the contract.

Daily Margin	The Mark-To-Market margin that a trader needs to provide to exchange due to daily settlement price changes.
Trading Method	It specifies if the contract can be settled as delivery of the underlying commodity or cash. Since the buyer of an Option has the right but not the obligation to exercise the right, the exchange specifies the Option's procedure.
Delivery/Settlement Terms	It specifies the terms of settlement of the contract.
Position Limits	It specifies the maximum number of contracts a trading entity can buy/sell. The position limit is vital as it does not allow a trading entity to corner the market.

Table: Typical contract terms of an Option contract

It specifies the terms of settlement of the contract. Since an Option contract is traded on an exchange, each exchange would have its terms defined for a contract. The terms and their definition can vary as per the exchange.

2.2.5 Swap Contract

A Swap is an agreement between two parties, where one party exchanges cash flow over an underlying/notional for a specific period. In the case of energy Swaps, one party agrees to exchange a fixed price of energy with a floating price or vice versa of an underlying

with a counterparty for a specific period. It can be a Swap between two floating legs. An example of a Swap can be an oil producer who will pay the average price of Western Texas Intermediate from the 21st to the 27th of the month and receive \$60 a barrel for 100000 barrels of oil every month for the next two years. The agreement just described is a Fix-Float Swap. Similarly, there can be a Float-Float Swap.

Swaps are generally used to hedge. Swaps are available for crude oil, gasoline, diesel, natural gas, heating oil, jet fuel, electricity, fuel oil, etc. Swaps are priced over an average price for a period mentioned in the contract. A Swap is not traded on exchanges but is Over-The-Counter trade. It is typically between hedgers and speculators or swap dealers. A Swap dealer gets into two deals where the hedgers want to offload the risk. The following diagram shows the Swap operations.

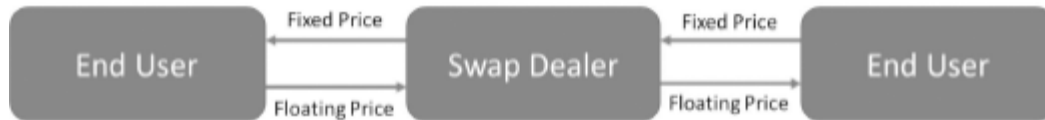


Figure: Swap Trade

2.2.6 Contract for Difference

Contract for Difference (CFD) – A commodity contract for difference (CFD) is a derivative instrument that mirrors the price movements of the commodity underlying the contract. CFD is used by speculators and retail investors mostly.

Thus, depending on the needs of a market player, different products are available to trade in Oil.

Chapter Summary

Comparison between Exchange and Over-The-Counter markets.

Parameter	Exchange	Over-The-Counter (OTC)
Customization	None. The contracts/securities traded are defined by the Exchange.	The contracts traded are 100% customizable to the needs of the parties.
Credit Risk	The exchange bears the risk involved in the execution of the trade.	The risk is borne by the parties involved in the trade.
Time	Exchanges operate on a fixed schedule, meaning an exchange has a set time to open and close.	24x7. It is dependent on the parties involved in the trade or the dealer making the trade.
Rules and Regulations	The parties involved need to adhere to the rules of the exchange. Moreover, they are governed by the regulations of the geography in which the exchange is located.	The trade is governed by the rules defined in the contract. There are light regulations in which the trade is getting executed.
Price Discovery	The prices for trading are provided by the exchange. The parties involved do not generally	The price discovery is done based on the negotiation of the parties involved. The parties know each other

	know each as trade matching is anonymous.	before they get into a trade.
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Table: Comparision between OTC and Exchange

The important products traded in the energy markets are Spot Trade, Forward Contract, Futures Contract, Options, Swap Contract and Contract for Difference.

TRADE LIFE CYCLE

From the topics that have been covered so far, we have understood the history of oil, the advantages of trading oil, types of markets and players. We have also understood the type of products that are traded. Now let us look at the entire oil trade life cycle, followed by trade settlement and the end-of-day process.

3.1 Trade Life Cycle

A trader in the oil trade is involved in buying and selling oil whether it is a hedger or speculator. Following is buying or selling process for a trader.



Figure: Basic trade lifecycle for a buy or a sell deal

As mentioned above this is a basic process. The process can be understood as follows:

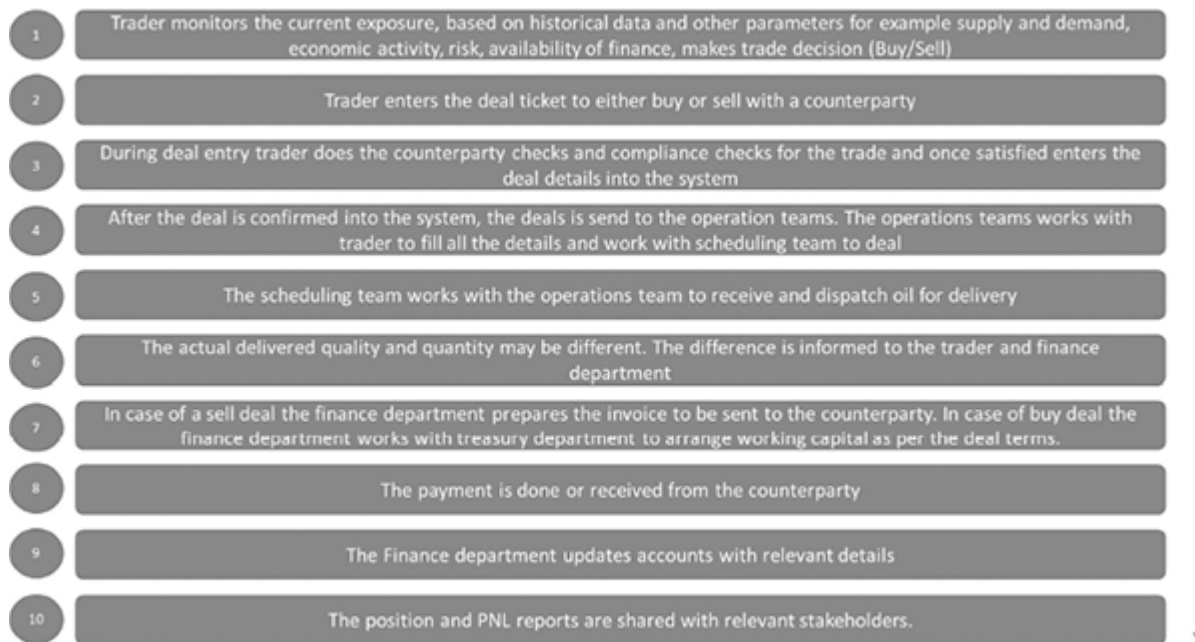


Figure: Basic steps in an oil trade life cycle

As mentioned in the previous sections, there are different types of players involved in oil trading. Oil trading involves risk, and hence a trading office is divided into the Front Office, Middle Office, and Back Office.

The corporate department includes the firm executive management. **Marketing department** whose responsibility includes converting firm ideas, products and or services into leads for the sales team. **The sales department** is the revenue generator of the company. **The public relations department** is essential in keeping the organization's reputation positive and **customer service** is responsible for developing relationships with current customers. All the departments mentioned are part of the Front Office of a typical organization.

The middle office includes departments like **the risk Management department** that identify risks and threats. **The compliance department** ensures compliance with regulations, rules, standards,

and specifications. The **Information Technology** department is involved in the management of hardware and software to run the organization. The **Financial Controls** department develops and implements policies, processes, and protocols used to manage the company's finances.

The **Human Resources** department overlooks employee performance, talent acquisition, and talent development to achieve firm goals. The **operations department** is responsible for executing the firm's core business processes. The **accounting department** is responsible for accurately recording the financial transactions taking place in the organization. The **manufacturing and logistics** department converts the raw material into products and ensures customer delivery. All these departments typically make up the **Back Office**.

The above classification is for a typical organization. Similarly, the trading organization is generally classified into the Front Office, Middle Office, and Back Office. The Front Office's primary functions are Sales Order Creation, Trader Order Management, Order Execution, Trade Execution, Portfolio Management, and Market research. Risk Management, Compliance, Performance and Analytics, Client Reporting, Regulatory Reporting, Fee Billing, Reconciliation Process, and OTC Derivatives Processing make up Middle Office. The Back Office includes Fund Accounting, Security Setup & Pricing, Trade Settlement, Reconciliation, and Fund Accounting.

The operations of an energy trading firm can be classified into Pre-Deal, Execute the Deal, Risk & Compliance and Manage Deal to Cash.

There are horizontal teams that support the trading organization.

The activities that are typically executed in Pre-Deal are Defining Strategy, Developing Go-To-Market, Identifying Customers, Onboard Customers, Sign Counterparty Agreement and Credit checks and setting up limits for the counterparty.

Executing the deal process involves evaluating the impact of the deals, conducting compliance checks, Pricing formula finalization, Deal capture, Cost capture, Trade confirmation, Book out and circling out deals, Link deals and Maintaining the portfolio.

The risk and compliance for deals are done by contract creation and signing, calculating market risk, calculating credit risk, monitoring limits, monitoring profit and loss, maintaining reference data, trade finance, forex risk management, and regulatory reporting.

Once the deal is done, The deal closure happens only when payment is revised or made. To achieve activities like reconciliation, managing collateral, charter and logistics, inventory management, demurrage calculation, dispute resolution, invoice processing, payment processing, business accounting and taxation.

Pre-Deal	Execute the deal	Risk and Compliance	Manage deal to Cash
Define Strategy	Evaluate the deal impact	Contract creation	Reconciliation
Develop Go to Market	Compliance check	Contract signing	Manage Control
Identify Customer	Pricing formula finalization	Calculate Market Risk	Charter and Logistics
Onboard Customer	Deal Capture	Calculate Credit Risk	Inventory Management
Sign up counterparty	Cost Capture	Monitor trading limits	Demurrage calculation
Credit checks	Trade confirmation	Monitor profit & loss	Dispute resolution
Limit's setup	Book out & Circle out deals	Maintain reference data	Invoice processing
	Link Deal	Trade Finance	Payment processing
	Maintain portfolio	Forex Risk Management	Business Accounting
		Regulatory Reporting	Taxation

Table: Activities during the trade life cycle

To perform the above activities, the trading organization setups up departments.

Pre-deal analysis, some of the prominent departments are Strategy, Know You Counterparty (KYC), Credit Analysts and Legal.

To **Execute the deal**, the organization needs Traders and Trade Operators.

To manage **Risk and Compliance**, the departments included are the Market Risk team, Credit Risk teams, Pricing and Market data team, Profit and Loss (PNL) team, Foreign exchange risk management team, and Regulatory Reporting team.

Supply chain management, accounting team, reporting team, invoice processing, payment processing, tax team, and trade processing team help convert **deals to cash**.

Pre-Deal	Execute the deal	Risk and Compliance	Manage deal to Cash
Strategy	Traders	Market Risk team	Trading Back Office
Know Your Counterparty (KYC)	Trade Operators	Credit Risk teams	Trade Finance
Credit Analysts		Pricing and Market data team	Supply chain management
Legal		Profit and Loss (PNL) team	Finance & Accounting team
		Foreign exchange risk management team	Invoice processing
		Regulatory Reporting team	Payment processing
			Tax team

Table: Teams during different the trade life cycles

The following sections describe the typical high-level functions of each department. The department in each trading organization may be set up differently in the way the organization feels that it would get the best competitive advantage.

Strategy Department: Every trading desk would have a view of the market. The view can be short-term and long-term. Material events keep happening worldwide, and it is difficult for a trader to keep an eye on every event. Thus, the strategy team comprises a variety of skills ranging from industry experts and economists to trading experts. The team prepares a short-term view and a long-term view of the market. The notes prepared by the team help the traders' direction. The strategy team subscribes to various data providers to

gather information. Currently, the team also has data scientists who would help develop mathematical models for trading.

Know Your Counterparty (KYC): A trader is always searching for a good trade with a counterparty. The trade can be with an existing counterparty or a new counterparty. From a compliance point of view, the first check is whether the counterparty party is a valid counterparty for the trading organization. The KYC team does maintenance of this record. In the case of a new counterparty, the KYC team gets in touch with the counterparty and collects relevant information to qualify the counterparty. The data can be audited balance sheets, counterparty tax details, and critical persons at the counterparty. The KYC team also checks for information from data providers to approve the counterparty. Moreover, the KYC team's other task is constantly monitoring the existing counterparties to ensure that their data is updated.

Credit Analysts: Nonperforming assets and credit defaults directly hit the trading organization's bottom line. To manage the risk, companies do deploy active and passive ways of credit risk management techniques. The techniques are costly to the trading organization and can affect the competitive advantage of the trading organization. Hence the team of credit analysts constantly evaluate the credit health of the counterparties. To do credit evaluation, credit analysts get credit data from rating agencies. One such data source is the probability of default of companies published by rating agencies. The team updates the credit limits for the counterparties. One of the challenges that the credit analyst team faces is a large number of counterparties that rating agencies do not cover Credit evaluation of

those counterparties has to be done in-house. Some trading organizations develop their internal ratings based on external rating and their assessment of counterparties. The work of credit analysts is to ensure that nonperforming assets and defaults remain minimum.

Legal Team: A large trading firm has general terms and conditions. One can visit the website and find these terms and conditions. The task of the legal team is to update the same regularly. Moreover, specific agreements are finalized with counterparties. The task of the legal team is to review the agreements and provide approval. The Over-the-Trade contract is also a legal document and needs legal review and approval. Trading is regulated by different geographies. The legal team's task is to ensure that the trading organization is legally compliant with the regulations.

Traders: The activities of a trader include analyzing the prices along with data regarding other parameters like supply and demand, weather, the economics of the different regions, and storage, to name a few and entering into a trade with the counterparty. A trader also needs to check the compliance of the trade before entering the trade. The trade should follow certain philosophies. For example, either it is to hedge a position or make a profit. Once the deal ticket is entered, the trader works with the trade operators to make sure the deal is closed. Traders also work closely with the strategy department to ensure the trade makes a profit. Managing the relationship with the counterparties and identifying new counterparties is also part of the role of a trader. Traders do work on specific products or specific

regions depending on their expertise. The ultimate aim of a trader is to make money.

If the strategy team and trader are working on wrong assumptions or strategies, then it can be catastrophic for the trading desk or the organization. Take the case of Meteallgesellschaft.

Meteallgesellschaft was incorporated in 1881 in Germany. As per the article of association, their main business was trading and manufacturing of metals and metal oxides. The company grew into one of the biggest German conglomerates. In the early 1990s, it decided to open an energy trading office in the US. The aim of the trading office was (a) to sell refined products in the Forward market and the physical market, (b) invest in refining capacity to produce the products and (c) Hedge Forward sales using derivatives.

The strategy was simple and might be used by hedgers. In 1992, current physical prices were lower than the Futures prices. Thus, the sales contracts were set at higher prices in the future. The purchase of a near-monthly contract was cheaper as the prices were lower. The strategy was to cover the sales of the far month by purchasing the near-month contract and rolling over on maturity. The strategy would create profit if the market kept the way it is.

The one problem with the strategy was cash flow management. As the payment of a long-term sales contract would be realized over the long term, if the prices increase in the near term, payment needs to be made for margin calls on the short-term contracts and more cash is required for rolling over the contracts.

In 1993, the same thing happened. The prices in the short term started rising. The market reversed from “Backwardation” to “Contango”, and management closed all the short-term positions. The closing of the contracts resulted in a loss of billions. Over a period in 1993, Meteallgesellschaft made a loss of \$1.5 Bn to \$ 2.2 Bn. The group had to ask for a bailout, and the conglomerate had to restructure. It has been debated that the group made a hasty decision to close the contracts which, otherwise was a sound trading strategy.

The above incident at Meteallgesellschaft showed that energy trading is not for the faint-hearted and is fraught with a loss without risk management or cash management. Moreover, oil trading is multi-dimensional and can quickly fluctuate between Backwardation and Contango.

Trade Operators: The success of a trade operator is measured in terms of no slippages that impact the profitability of the trade. Operators are a vital part of an energy trading organization. The primary role of the operator is to make sure that the deal fructifies as intended by the trader. The first task for a trade operator is to work closely with the trader and ensure all the trade details are captured correctly. Trade operators work closely with the traders and most other functions mentioned in the above table. A trade operator should have versatile knowledge involving chartering, supply chain, legal, and risk management taxation, to name a few. Trade operators build a strong relationship with internal and external (counterparties, terminal operators, inspectors etc.) to ensure the deal is executed (oil is delivered or received, and payment is received or paid). They

constantly monitor the deal's PnL and alert the trader and other stakeholders if there is a mismatch between the expected actual and projected PnL.

Credit Risk team – One of the significant concerns of the trading organization is counterparty default. Again most of the counterparties in energy trading are not covered by rating agencies. Hence the function of the credit risk team becomes critical. The credit risk team primarily manages the Mark-To-Market (MTM) and Settlement Risk for an active deal. The team is also required to models and processes to mitigate credit risk. They also create margin calls on a deal with the counterparty and issue margin calls to the counterparties. They ensure all the credit risk management processes are followed. The credit risk team is responsible for producing credit reports for all the relevant stakeholders. They are also responsible for limits management on the positions. The credit risk is measured on a trade, portfolio, and for an entire trading organization—the credit risk team also participates in the purchase of insurance to cover the credit default risk. The importance of credit risk has increased as banks have tightened trade finance requirements.

Market Risk team – The price volatility is why different players enter energy trading. The volatility in prices raises the risk of loss. The market team is responsible for managing market risk. The market risk team calculate risks for a trade, portfolio or trading organization. The most common method used by the market risk team is Value-at-Risk. The other activities include monitoring the limits for market risk. The team will also produce reports for market risk to various stakeholders in the trading organization. The market risk team is also responsible

for helping reduce the market risk and advising on hedging strategies. Since commodities are one of the most volatile calculating, monitoring, reporting, and managing market risk is a vital activity as part of the risk management team.

Pricing and Market Data team – The trading organization needs data to trade. The data might include pricing data and market data. There are data service providers that provide benchmark data and pricing data from across the globe. Some leading market data providers are S&P Global Platts, Refinitiv, HIS Markit, Morning Star, and Bloomberg, to name a few. There are so many different grades, so it is difficult for all the market data providers to provide prices. So a broker also provides data. One of the tasks of the pricing and market data team is to build pricing curves from the data available. The vital task for the team is to provide a golden data source. All the stakeholders use the data produced by the team. For example, the traders would use the data to price the deal, the risk team would use the data to calculate risk, and the back office team would use the data for trade confirmation with the counterparty. Thus, the pricing and data team would have a vital function to be the data owner based on the entire trading organization's operation.

Profit and Loss team – Different organizations, might have another name for the team, but the team's primary function is to calculate the profit and loss for the deal. Two types of profits exist in a deal. The first is called the notional or theoretical profit, and the second is the actual profit. The team's responsibility is to ensure the calculation of the profit and loss is correct. The team is also responsible for reporting the profit and loss on a trade, portfolio, desk or entire

trading organization. Some of the vital tracking points are profit for the day, profit for the month, profit for the quarter and profit for the team. The team produces a PnL report daily and circulates the same with various stakeholders.

Foreign Exchange (Forex) management team – Crude oil is priced in dollars. The geography in which the trading desk or trading organization operates might differ, and the reporting currency might be different. As a result, most trading organizations worldwide have a team that manages the foreign exchange risk and requirements resulting from trading. In a vertically integrated organization, forex needs and risk have to be managed actively, and hence the role of the team becomes essential. The team's primary aim is to manage foreign exchange risk that arises out of the trading positions.

Regulatory reporting team – Compliance is one of the critical functions of a trading organization. The legal team helps the trading organization to be compliant with local regulations. Now local jurisdictions have reporting requirements. Worldwide there are different regulations, and some of them have reporting requirements. Some of the regulations that govern energy trading are the Market Abuse Regulation (MAR and MAD), the European Market Infrastructure Regulation (EMIR), the Markets in Financial Instruments Directive and Regulation (MiFID II and MiFIR), Dodd and Frank, to name a few. Some of the regulations have mandatory reporting requirements. Some of the reporting needs to be done daily or at regular time intervals. The task of the regulatory reporting team

is to ensure that the reports get prepared and submitted to the regulator as per the regulation.

Trading Back Office – A trade reconciliation process ensures no dispute due to differences in trade booked by the trading organization and brokers for the trades executed via brokers. The reconciliation process makes sure the transactions are booked correctly. Trade confirmation happens once the deal is agreed upon. Trade affirmation is done if the trade is made directly with the counterparties. Electronic trade confirmations are done with the relevant stakeholders for transactions done via a broker or exchange. The back office team ensures the prices are updated correctly on the invoices and other documents. As part of the post-settlement process, the following processes depend on whether it is financially or physically settled. The team is responsible for updating the deal with all the relevant data to ensure that other teams can work correctly on the deal. For example, updating the accurate prices ensures that the invoice process team can create correct invoices. In summary, the back office team is responsible for reconciliation, affirmations, confirmations, updating pricing and other details, and trade settlement.

Sometimes the shipping insurance team is part of the trading back office, and they work with the insuring companies and ensure, depending on the shipping terms, insure the cargo.

Trade Finance – To complete a trade, one of the instruments used is a Letter of Credit. Most of the physical deals need to be covered by a Letter of Credit to make and receive payment. The primary function

of the trade finance team is to manage the letter of credit for the deals, whether it is a buy or a sell deal. The trade finance team works closely with the bank to issue a letter of credit or accept a letter of credit. The other function of the trade finance team is to discount the letter of credit to manage the finance of the trading desk. The value of the letter of credit is high as the volume of oil traded in one trade is huge; thus, getting a good quote on a letter of credit directly impacts the profitability of the trade.

Supply Chain Management – There are two types of deals: the physically settled deal and the financially settled deal. The physically settled deal is a supply chain to ensure that the oil is delivered. It is also required for oil to be received. One of the primary functions of the supply chain team is to make sure the vessel is available to deliver the quantity of oil mentioned in the trade. The team is also responsible for ensuring the correct quality is delivered or received. To ensure this, they need to work with inspectors involved with “Crude Assay”. They need to work with the customs department or custom house agents to ensure there is no hindrance. If the oil needs to be stored, then the storage should be available. If there is a difference in crude quality, then the supply chain team has to go through the dispute resolution process as per the contract. They have to make sure all the documents are dispatched and received correctly. They also have to approve the invoices of the third party involved in the process. They have to maintain a relationship with all the stakeholders. Lastly, they have to ensure that the invoices have correct quality, quantity and value as a minimum. If the trading organization is involved with freight trading, they need to work closely with the team

to register the interest. The operations differ slightly in the case of physically settled exchange trade. The supply chain has to work with the exchange and the clearinghouse of the exchanges to fulfil their obligations. In summary, supply chain management is critical to oil trading when the delivery is physical.

Finance and Accounting (F&A) – Different types of deals have different payment conditions. The responsibility of the finance team is to ensure that the payment conditions are adhered to. The treasury team, which is part of the finance team, has to ensure that all the appropriate capital management team work is done. The team that works with the trade finance team described above can be part of the finance team. The finance team is responsible for financial reporting as part of fiduciary responsibilities. There are liabilities due to trading and finance, and the accounting team has to be managed actively. There are two types of profits, one being theoretical and one being actual profit. As in the F&A, different types of deals have different payment conditions. The responsibility of the finance team is to ensure that the payment conditions have been adhered to. The treasury team, part of the finance team, has to ensure that all the appropriate capital management team needs are done. The team that works with the trade finance team described above can be part of the finance team. The finance team is responsible for financial reporting as part of fiduciary responsibilities. There are liabilities due to trading and finance, and the accounting team has to be managed actively. There are two types of profits, one being theoretical and one being actual profit. As the F&A team, it is their responsibility to record the entries correctly. The team is also responsible for working with rating

agencies to get the required credit agencies. The F&A team is the backbone of the entire trading organization. The team is responsible for recording the entries correctly. The team is also responsible for working with rating agencies to get the required credit agencies. The F&A team is the backbone of the entire trading organization.

Invoice and Payment Processing – The value of each deal is high in the case of oil trades, and hence it becomes essential that correct invoices are generated. Moreover, the terms of payments are stringent and thus must generate an accurate invoice. Moreover, as part of the invoice processing, the team will need to process an invoice from the deal, agents, inspectors, shipping department, and insurance team, to name a few. All invoices need to be processed for accuracy. Also, the invoice team needs to generate invoices for the sell deals.

Similarly, the payment processing team is responsible for making the payments after the invoices are processed. They are also responsible for working with the counterparties to ensure that the payment is received. The teamwork with both internal and external stakeholders to provide proper documentation and payment is processed.

Taxation – Trading deals happen over multiple taxation jurisdictions. A taxation team ensures that tax liabilities are met for all the deals and the trading organization.

3.2 Trade Settlement

One of the crucial activities of the back office of a trading desk is trade settlement. With the trade settlement, the deal is concluded. There are two modes of settlement possible in the case of derivatives trading in Energy trading. The first mode is the cash settlement. The second

mode of settlement is the physical settlement. The type of settlement is already mentioned in the contract irrespective of OTC or Exchange trade. In the following two sections, we will detail the two modes of settlement.

3.2.1 Cash Settlement

Cash settlement is a mode in which the seller in the trade agrees to a net cash position for the trade rather than the physical delivery. In the following example of future trade, the details of the future traded on the exchange specify that it will be a financial settlement. So, if the Futures contract is for 1000 barrels, it must be settled financially. If the buyer has purchased the contract at \$100 a barrel on the day of the settlement, if the settlement price of the oil published by the exchange is \$110 a barrel, then the buyer will get a settlement of \$1000 for the Futures contract. Similarly, if the settlement price of the barrel of oil was published as \$90 for the Futures contract, the buyer will have to pay \$1000 for settlement.

The above example is to explain the cash settlement of the contract. Similarly, other derivatives like Forwards, Options or swaps can be financially settled. In the case of an Exchange, different mechanisms like initial margin, maintenance margin, and other margins must be exchanged during the entire trade lifecycle to eliminate credit risk. Risk management is of paramount importance, and various techniques are implemented for risk calculation and control.

3.3.2 Physical Settlement

In the physical mode, the seller must deliver the physical delivery of the underlying. The trade description would describe the trade would

be settled by physical delivery. The trade would include a description that mentions the delivery's quality and location in the trade contract/trade.

The role of the exchange is vital as it will match the buyer and seller. Also, ensure that the seller and buyer meet the terms and no credit risk exists. One of the other essential things is that the seller has to inform the exchange that they are ready to deliver the commodity. For example, a buyer of the Forward contract will get an intimation of delivering the crude oil from the seller as the settlement date arrives. Depending on the terms of the trade, the seller would ensure the crude oil is delivered. In the case of exchange trade, the seller of the future would inform the exchange about the intention to deliver. The exchange would facilitate the settlement, including the clearinghouse.

One of the things to remember in the case of energy trading is that the delivery of the quantity and quality of the underlying might not be exact. Hence, there needs to be a mechanism to accommodate the variances. Moreover, there can be a dispute in terms of quality, and a solution mechanism needs to be in place in case of an OTC trade. The exchange has its mechanisms in place to deal with such deviations.

The following table shows a brief comparison of financially and physically settled trades

Basis of Comparison	Financially Settled	Physically Settled
Liquidity	Higher than Physically Settled.	Lower than Financially settled.
Settlement time	Generally, it is instantly.	Long and payment might include a

		credit period.
Cost	None other than the cost to trade on the exchange.	Higher delivery costs, fees and other costs might be included.
Risk	None in the case of Exchange-traded.	None in the case of Exchange-traded.
Convenience/Popularity	High, compared to physically settled.	Low as done primarily by consumers and producers.

Table: Comparision between Physically and Financially settled trades

3.3 End-of-Day (EOD) Process

Exchange trades happen during the time exchange is open. While OTC is available 24x7, thus, it is possible to trade 24x7. A trading desk works mainly in the local times. Now imagine when the traders, risk analysts and back-office analysts come to the office and would like to know the value of their portfolio, market to market, profit and loss, etc., as a starting position to make effective trading decisions during the day.

To make the above requirement of different offices possible, the trading organisation runs a process called the end-of-the-day process. The following section provides a high-level understanding of the

process, some organisations might change the order of the process, but the outcome at the end of the day process remains the same.

During the trading day, the prices fluctuate. To calculate an accurate position, the pricing team has to publish the prices at which all the trades open and at the end of trading, it will be valued. The publication of prices is called the End-of-Day prices or end-of-the-day curve data.

There can be multiple applications or systems on which trading happens during the day. For example, the OTC trades are booked to custom-build applications or ETRM packages installed by the trading organisation. Trading on the exchange might be done on the trading terminals provided by the exchanges. Some of the trades might be with brokerages. Some of the trades might have been settled during the day. So, one of the activities done at the start of the end-of-the-day process is a consolidation of all the trades in their respective portfolios.

The prices/curve data is available to price the trades. The trades that need to be valued are available to calculate the portfolio position, Mark-To-Market, value at risk, profit and loss, etc.

One of the activities is to start trade valuations. If some trades are missing, the support team intervenes or attributes reasons for the same. After correcting the cause, the trade valuation is done. The outcome of the process is proper valuation of the trades. Along with the trade valuation, profit and loss are calculated for the trades, portfolio or desk depending on the organisation.

After trade valuation, the trades are used to calculate market risk. For the calculation of the market, a risk engine is used to help calculate the Value at Risk. Depending on the risk philosophy of the company, various reports are created. If there are VaR (Value at Risk) limits available, then checking against limits as part of risk management would be done. The reports would be ready for the market risk team in the middle office for action the next day.

Similarly, the credit risk engine calculates the credit risk for the trades. A couple of essential parameters that are calculated are Mark-To-Market and settlement risk. The two parameters are crucial as, based on the report, the team would make or honour a margin call. The reports are essential for the treasury team as these would directly import the cash flow.

Apart from trade valuation, profit and loss, market risk and credit risk, many other reports regarding corporate compliance reporting are produced.

Most organisations have trading desks worldwide, so they use the “follow the sun” model for the EOD process. For example, a trading organisation has a trading desk in London, Chicago, Houston, Tokyo, Singapore and Dubai. Thus, the trading day for the trading organisation would end in Houston. The trading day would start in Japan. So, the trading organisation would try to utilise the time between Houston closing and Tokyo opening for their End-Of-Day process.

An efficient EOD process management can give an organisation a competitive edge. Since the subsequent day trading depends on the

successful completion of the End of Day process, the trading organisation invests in making the process accurate, automated and robust. As seen in the example above, the time available to complete the function is short, and hence the trading desk invests in computational power.

Chapter Summary

A trade life cycle helps us understand different activities that need to be performed to complete a trade

The front office is where the trading happens. The middle office is responsible for Risk and Profit & Loss calculation and Management. The back office ensures trade fulfilment.

The oil trading desk is like a bee hive in terms of the activities to be performed to complete a trade. All the teams need specialised skills to operate in their area. All the activities are there to enable the trader who is at the centre to make the right trade.

RISK AND RISK MANAGEMENT

The four major trading asset classes look at Equity, Interest rate, Currency and Commodities. Historically Commodities have high volatility. The quarterly crude volatility since 1983 is between 12.63% to 90%. The range for natural gas in between is from 22.56% to 80%. The ranges are the same for sugar, corn, silver, or gold, if not on the lower side. Compared to this, S&P 500 volatility has been around 5.35% to 27.23%. Bond volatility has been from 6.22% to 17.5%, and the same has increased since 2008. For currencies, let us look at Dollar Index, and its volatility range is between 4.37% to 15%. Thus, making commodities the most volatile asset class.

We also know that oil trading is multi-dimensional. Hence risk calculation and risk management are of paramount importance. Once we know the risk, we can do risk management, and hence that is the last topic of the chapter. The subsequent sections will help understand the Market Risk, Credit Risk, Concentration Risk, some other risks and risk management.

4.1 Market Risk

The volatility in the markets is one reason that measuring and managing price risk is paramount. Price risk (a.k.a market risk) is defined as the possibility of risk of losses in position arising from movements in market variables like price and volatility. One of the characteristics of market risk is that always present. The price keeps changing, and hence the risk is always there. Hence, as mentioned

above, actively measuring and taking action is a way to manage market risk.

The way to measure the market risk is to use a statistical method called Value at Risk (VaR). Value at Risk (VaR) is defined as the maximum potential loss for a portfolio given a specific confidence interval over a specified period. For example, one day VaR at a 95% confidence level is \$1 million for a given portfolio means that tomorrow there is 95% confidence that the maximum loss on the given portfolio due to adverse price change is \$1 million. Similarly, one day VaR is at a 99% confidence level is \$500K for a given portfolio is understood as that tomorrow there is 99% confidence that the maximum loss on the given portfolio due to adverse price change is \$500k. From the example, one can notice that VaR model results are simple for all levels of staff from all areas of an organization to understand and appreciate. That is why VaR is a widely adopted method for measuring market risk.

There are different methods to measure VaR. The three methods prevalent to measure VaR are (a) Analytical, (b) Historical and (c) Monte Carlo Simulation.

Parametric Method: The primary method parametric method used is Variance-Covariance. IT is an easy and fast calculating method to calculate VAR. The method is preferred as volatilities are predictable. The best use is for currencies and financial instruments. Returns of the underlying should have a normal distribution for using the method. The parametric method is not used for a portfolio with many Options as Options are not linear.

Historical Method: Historical simulation method is a simple method with easy mathematical calculation, without necessary statistical parameters. The method captures the real distributions of the factors as no assumptions are made. The method is easy to explain to others. The Basel Committee recommends at least one year of historical data points to consider when using the Historical Method to calculate VaR. One of the disadvantages of this method is that there can be a significant difference between the past and future distribution factors. The other disadvantage is that all the factors in the past are equal-weighted. Finally, selecting a historical period for the input becomes critical, and an incorrect choice might lead to undesired results.

Monte Carlo Simulation Method: The method is flexible and can generate a large number of reliable data that can be used for the calculation of VaR. The model captures parameters (like convexity), making it suitable for non-linear products and making a method of choice to calculate VaR for a portfolio having multiple types of products. The drawback of this method is a lot of computational power is required to run the simulation. The other point to remember is that while using this method is that the distribution is an assumption and might differ from the actual. Overall, if there is more than one risk variable affecting the outcome, the Monte Carlo simulation is preferred.

The above-mentioned three methods are the primary methods for calculating VaR. The other methods to calculate VaR are:

Extreme Value Theory (EVT): The EVT is based on extreme events, including significant losses rather than small profits. The theory is

often used when estimating risk in crises due to the extreme variations in markets.

The **Conditional Autoregressive Value at Risk** model, also often referred to as the CAViaR model, focus on the quartile rather than the distribution of the return of the instrument.

The **Exponentially Weighted Moving Average (EWMA)** theory is about weighing together all historical prices of an asset, where the more recent prices are given greater weights to constitute their importance. This is because an assumption is made stating prices closer to today be more relevant for the outcome of tomorrow's price.

Orthogonal GARCH (O-GARCH) model, data obtained from linear transformations are used for estimating VaR. The O-GARCH is based on unconditionally uncorrelated variables taken from original data.

VaR has its limitations, and the limits of VaR need to be kept in mind while using VaR. Some of the limitations of VaR have been summarised as follows:

Huge portfolios: The calculation of VAR for a portfolio requires one to calculate the risk and return of each asset along with the correlations between them. Hence greater the number or diversity of assets in a portfolio, the more difficult it is to calculate VaR.

Different methods: The choice of method depends on the nature of the portfolio. Different approaches to calculating VaR can lead to different results for the same portfolio.

Assumptions: Calculation of VaR requires one to make some assumptions and use them as inputs. If the premises are not valid, then neither is the VaR figure.

The following chart compares the different methods of VaR on various parameters:

Parameters	Parametric Method (Variance/ Covariance)	Historical Method	Monte Carlo Simulation Method
Implementation	Easy to implement because it assumes a linear distribution.	Relatively easy to implement and Implementation ease falls in between the Parametric and Monte Carlo Simulation.	A huge number of iterations will have to be carried out to get a reasonable result and hence it is hard to implement.
Accuracy	The variance-covariance method has a very high accuracy.	The historical simulation method is not very accurate when a short period is considered.	Monte-Carlo method predicts risk with high accuracy for a large number of iterations.
Product Limitations	Non-linear instruments (Ex: Option) cannot be accounted for by this method.	This method cannot account for events like spikes in risk factors (unusual events) and the variation in risk due to time.	May be very expensive and time-consuming to build the simulation. The major pitfall is that of running a risk with the financial model.
Speed	Computations can be performed quickly.	Computations can be performed quickly.	Computation is slow when compared to the other two methods.
In times of Extreme Events	As the distribution of risk factors is normal in the Parametric Method the probability of extreme events happening is very minimal.	The possibility of extreme events happening is only more relevant if they happened in recent history.	The Monte-Carlo Simulation method due to its completely random nature accounts for these events completely.

Table: Comparision between different methods to calculate VAR

The methods mentioned above help to calculate VaR. The underlying asset can be a financial instrument based on a derivative's price; it could be a currency, a commodity, a stock, or a bond. For regulatory requirements, the calculation method is decided by geographies. One concluding point is that measuring risk is the start of risk management and not the end.

4.2 Credit Risk

Enron collapsed in 2001, followed by the bankruptcy of other energy firms like TXU Europe and AES Drax, which led to a situation where suddenly 70 large banks servicing the energy sector were reduced to only a dozen banks willing to serve the energy sector. 2001 was the first credit squeeze that happened for energy firms.

Later during the financial crisis from 2008-2009, oil prices fell from a high of \$147 in July 2008 to a low of \$33 in February 2009. In the absence of credit support, many firms had to increase capital. The lesson from the crisis was to improve credit risk assessment techniques.

Again, in 2015, with the plunge in commodities prices, the industry saw a sudden increase in the price of Credit Default Swaps for Glencore. Similar Noble Energy got its bonds rated to junk status in Dec-15 after an initial adverse report in Feb 2015.

In 2018, with the fluctuations in commodities, prices fluctuated and in April-May 2020, a scenario as never seen before unfolded. The future prices turned negative for West Texas Intermediate adding a new dimension to price fluctuations. Due to such volatile price

fluctuations, the cash flow of oil and gas companies comes under pressure. A profitable company without mismanaged cash flow can go bankrupt. Thus, oil and gas companies started focussing on Credit Risk. The buyer and supplier do not want to end up with credit losses, and hence the focus will only go further in this area.

The traditional definition of credit risk is the possibility of a loss resulting from a borrower's failure to repay a loan or meet contractual obligations. From an energy trading point of view, the term—"meet contractual obligations" gives rise to credit risk. For example, the upstream O&G company has an agreement to sell 100000 barrels of oil to a refinery. The contract provides 60 days credit period (The credit period is the number of days that a customer can wait before paying an invoice). After the oil is delivered by the upstream O&G company to the refinery, during the credit period, if the refinery goes into bankruptcy, the upstream company does not get paid and has to incur the loss. The reason for the loss is that the refinery before "meeting contractual obligations" went bankrupt. Thus, credit risk becomes one of the important parameters to measure.

Just as for market (price) risk we use VAR to measure the risk, for credit risk we use two measures. The two measures are (a) Mark-To-Market and (b) Settlement Risk. Many organizations term Settlement Risk in their own terminologies but to keep simple we would use these two terminologies.

4.2.1 Mark-to-Market

Mark-to-market is an accounting method used for risk management. Mark-to-market can be understood as a method of measuring the fair

value of accounts that can fluctuate over time, such as assets and liabilities. Mark-to-market aims to provide a realistic appraisal of an institution's or company's current financial situation based on current market conditions.

To understand why there is a need to measure mark-to-market (or MTM as it is known), let us go through the following example.

Company X, an upstream company producing crude oil, has decided to sell 100,000 barrels of oil to Company A, a refinery at \$50 a barrel to be delivered after two months.

The day after the agreement has been reached at \$49 a barrel, and if the refinery Company A files for bankruptcy. What would happen to the contract?

Company X must find a new buyer. The contract would be signed at \$49 a barrel, thus incurring a \$1 per barrel loss.

Thus, the total loss incurred by company X is \$100,000 as company A has filed for bankruptcy.

The above scenario is also true for the buyer where the supplier files for bankruptcy. In such a scenario, the buyer must find a new supplier. One can conclude that MTM can be positive or negative, i.e., depending on the market movement, the buyer/seller can profit if the counterparty files for bankruptcy. Since filing for bankruptcy is a credit event, the measuring of MTM falls under credit risk.

Credit risk is measured by the company (be it buyer or supplier) daily until delivery is not made.

4.2.2 Settlement Risk

Settlement risk is the possibility that one or more parties will fail to deliver on the terms of a contract at the agreed-upon time. For example, you decide to prepare an assignment for your friend when your friend offers to buy dinner on completion of the job. Now, if you fail to prepare the assignment, then your friend will get into trouble. If you finish the assignment and the friend does not buy you dinner, then your effort is wasted. In both scenarios, either you or your friend failed to deliver on the agreed terms and this is an example of settlement risk.

Now to understand Settlement Risk, let us extend the terms of the deal from the above section.

Company X, an upstream company producing crude oil, has decided to Sell 100000 barrels of oil to Company A, a refinery at \$50 a barrel to be delivered after two months with a credit period of 30 days. This means that company A has to make an oil payment to company X after 30 days (credit period).

Now the day after receiving the shipment, imagine the refinery filing for bankruptcy. Then company X does not get any money for shipment made to Company A.

The loss that Company X have to bear is $\$50 \times 100000 = \5000000 .

Thus once the goods are delivered, and the payment is received, the counterparties (buyer and seller) both measure settlement risk. We say both is the buyer has to manage cash flow to ensure payment is made at the end of the credit period.

Since the calculation of MTM and Settlement is dependent on ownership of the goods, one can refer to incoterms to identify what

needs to be calculated for the deal which has not been completed (completed means the commodity has been delivered and payment has been made). The following table summarizes the point during the deal lifecycle to determine the risk parameter (MTM & Settlement) that needs to be monitored. The point to note is that cost and risk transfer are different.

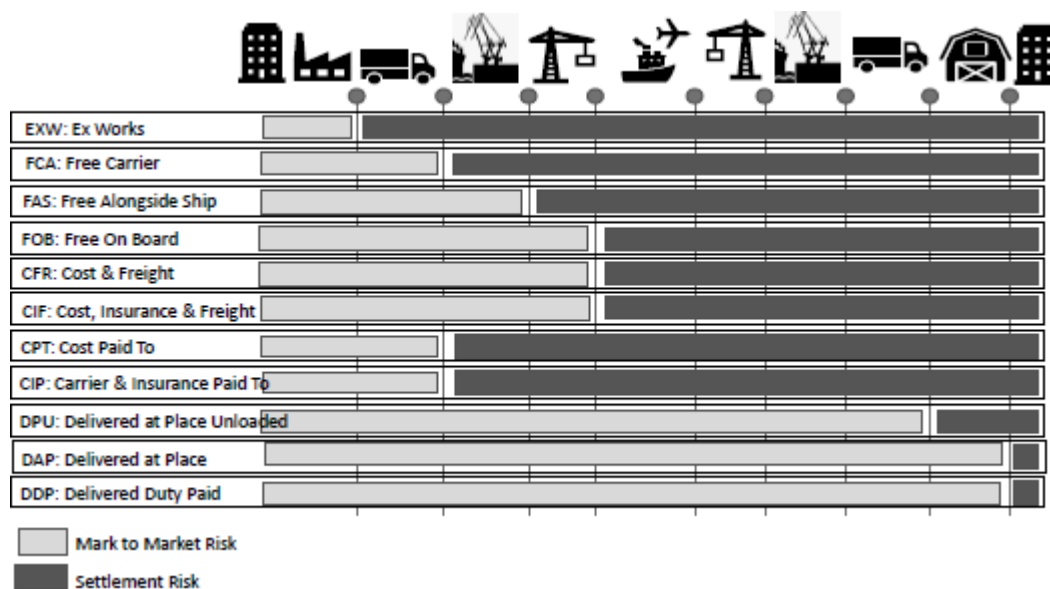


Figure: INCO Terms and Risk

4.3 Concentration Risk

Concentration risk is defined as a risk arising from concentration on a single counterparty, sector, or country. Oil trading is multidimensional. Oil production is limited to certain countries; refineries are concentrated in specific regions, and storage locations are in different areas of the world, sometimes owned by governments in terms of strategic reserves. Moreover, there are different sizes of counterparties involved in energy trading. Some companies are part of fortune 100 or 500, privately held trading companies, and small companies that are either into exploration and production or refining.

As a result, a trading desk needs to calculate the concentration risk for a portfolio, particularly in terms of country, counterparty, and banks (this would be clearer during risk management). For example, if, as a trading desk, most of your counterparties are in China, then any change in the business dynamics in China would impact all the counterparties. Hence a trading desk would always calculate concentration risk for a country. Sometimes the risk is called “Country Risk”.

Like China is the biggest trading partner of the United States (US), whatever happens in China impacts the US. Similarly, suppose Shell is making maximum trades with British Petroleum, and an event like the Gulf of Mexico happen. In that case, it is not just British Petroleum facing the risk, but as the most significant counterparty of Shell, it will also face colossal risk. Thus, the trading desk measures the concentration risk for the counterparty in terms of examples.

In oil trading, one of the tools of trade finance is a Letter of Credit (LC). Imagine if you are trading on behalf of Shell and all the counterparties start issuing LCs from Bank of America. Then there is a concentration of LCs from Bank of America. Imagine if this were in 2008 during the global financial crisis, then the Shell trading desk would have to worry about the financial health of Bank of America. Thus, trading desks do measure the concentration risk in terms of the collaterals they receive.

Oil trading is multi-dimension, so concentration risk becomes one of the essential risks monitored by the trading desk.

4.4 Other Risks

4.4.1 Operational Risk

Various regulators define operational risk differently across the world. For a beginner, the simplest definition is the uncertainties a company faces while carrying out daily operations/activities. In the case of energy trading, some activities like manual trade entry, back office settlement error, logistics or trade control failures are sources of operational risks. Following is one of the recent stories of the realization of operational risk at scale.

On June 30, 2009, a drunk trader working for a leading oil broker worldwide had 7,000,000 barrels of oil valued at approximately USD 520 million in two and half hours. The trader was not a rookie but had been with the firm since 1998. On June 29, after spending a weekend drinking heavily at a company-sponsored golf weekend and continuing to drink through midday and beginning at 1:22 A.M. on June 30, the trader started making the trades while still blacked out. The trading stopped at 3:41 A.M., approximately two and a half hours later. On contacting an assistant clerk enquiring about the trades, the trader was informed that it was on behalf of a client.

The volume was so massive that it represented 69% of the volume of oil being traded at that time. The impact was that worldwide oil future prices rose from \$71.40 per barrel to \$73.05 per barrel.

The financial service authority barred the trader from trading for five years and imposed a penalty of £ 150,000, which was reduced to £72,000, to be paid in 36 instalments.

The incident highlights the operational risks involved and focuses on controls placed by the firms on trading. Looking at the rise in the

price of crude, one cannot imagine the impact on the world when traders go recklessly.

Operational risks are systematic risks and hence the organization should be vigilant across the same.

4.4.2 Political Risk

The risk faced by a trading organization due to political decisions, events or conditions that impact profitability is called Political Risk. In a broader definition, one can cover changes in governments, trade tensions, sanctions, tariffs and political violence.

According to one study, in 2022 the political risk is nearly at the highest level. Political risk can be considered at two levels. Macro-level political risk is not specific to a project or an entity or a project in the country but affects all the participants in a given country. For oil trading, the US and European Union put sanctions on Venezuela. The sanctions were broad and impacted individuals and companies. Though the sanctions were not imposed by the UN major countries in North America and Europe passed resolutions and executive orders in 2015, for oil it came in 2017. Similarly, the UN imposed sanctions on Iran making it nearly impossible to do free trade in oil with Iran. After Russia invaded Ukraine, there were sanctions imposed on Russia and the oil and gas trade was highly monitored and regulated. Countries like the US and UK imposed sanctions on the import of Russian oil and gas. Thus, macro-level political risk impacts all the participants at large.

Sanctions that target projects or companies are considered an example of a micro level of political risk. The US imposed sanctions on five

companies (Petro Grand FZE, Alphabet International DMCC, Swissol Trade DMCC, Alam Althrwah General Trading LLC and Alwaneo LLC Co) in 2019 as these companies were involved in trading oil with Iran.

4.5 Risk Management

There are different approaches to Risk Management in oil trading. Various trading organizations have different risk appetites. Following are some of the techniques that are generally observed in the industry. Companies can deploy one or more of the methods for risk management. Company-specific innovative risk management is the Intellectual property of those companies and hence not described as follows.

Risk management can be broadly classified as risk management attributed to a counterparty, i.e., external to the trading desk. The trading organization can deploy a technique to manage the risk internally without interacting with the counterparty and can be considered internal.

Some of the techniques attributed to counterparty are (a) Netting, (b) Trade Finance and (c) Collaterals. The ones attributed internally to the trading organization are (a) Limits and (b) Insurance.

4.5.1 Netting

Netting is a method of reducing risk by offsetting trade or a portfolio with a counterparty. For example, if Shell has a trade to buy 100000 barrels of Oil from British Petroleum. There is another trade in the portfolio where Shell is selling 200000 barrels of Oil to British Petroleum. Thus, the total exposure of Shell to British Petroleum is selling 100000 barrels of Oil. To implement netting as a risk

management technique, the counterparties sign agreement with counterparties to allow netting. Netting does not have a cost as it is a mathematical calculation. Netting helps manage the credit risk of the trading organization.

4.5.2 Trade Finance (Letter of Credit)

A letter of credit is an instrument issued by a buyer to the seller for a purchase order. In the case of oil trading, imagine Shell is buying 100000 barrels of oil from Trafigura. So as a buyer, if Shell issues a Letter of Credit to Trading for the buy contract, then while calculating risk, Trafigura would reduce the risk exposure value by the value of the letter of credit received from Shell. To issue a letter of credit, the buyer must pay fees to the bank, and thus pricing of the trade would be different. Since the letter of credit is a sacrosanct document it is actively used for risk management. But as mentioned above, a letter of credit issued by the same bank increase concentration risk for the trading organization. Trade finance reduces settlement risk for a trading organization.

4.5.3 Collaterals

The trading organization can ask for collaterals from the counterparty to reduce the exposure and risk. The collateral that a trading organization can request can be cash or bonds. The exact amount can be used as part of the trade settlement process. Since the counterparty is provided collateral, the same is added to the trading cost or cost of landed or sold products. Collaterals are used to manage the credit risk of the trading organization.

4.5.4 Insurance

If a trading organization fills that, there is a risk of counterparty default. Another reason can be a risk to trading with a counterparty due to uncertainty around the counterparty in terms of regulations (sanctions) or the reputation of trading with the counterparty. For a similar scenario, the trading organization buys insurance against the losses arising out of unforeseen circumstances. The insurance purchased by the trading organization is not disclosed to the counterparty. The cost of insurance is part of the cost of trade and deducted from the profit of the trade or included in the landed price or selling price, depending on the type of trade. Buying insurance is to manage the credit risk of the trading organization.

4.5.5 Limits

As mentioned, multiple times, different trading organizations have different types of risk appetite. As risk is involved in trading, the trading organization put a different limit to the overall exposure or overall position of the trading organization as a whole or against a counterparty or for a country or any other risk parameters. Limits are majorly used in credit risk management. For example, a trading organization would put a limit on Mark-To-Market with a counterparty. If a general term and condition signed were beyond the Mark-To-Market, the trading organization would place a margin call. Similarly, a trading organization would limit overall exposure to a counterparty in terms of the trading volume. The other way limits are set on the settlement amount of the counterparty, and if the counterparty exceeds the limits, then pre-payment can be requested by the trading organization.

The other place limits are used effectively to manage concentration risk for a country or a bank. The trading limits in both scenarios are set against the overall exposure to a country or a bank. If the limit is breached, then trading is stopped for counterparties registered in the counterparty. In the case of banks, the trading organization would stop accepting collaterals or letters of credit issued by the bank.

In the case of market risk, limits can be put against the value at risk amount at different levels. For example, limits are set on value at risk for a portfolio. If the limit is breached, then the trading organization would take hedging positions and reduce their value at risk.

Trading organizations do deploy advanced techniques as part of risk management.

Chapter Summary

Market Risk: Price risk (a.k.a Market risk) is defined as the possibility of risk of losses in position arising from movements in market variables like price and volatility.

Credit Risk: The traditional definition of credit risk is the possibility of a loss resulting from a borrower's failure to repay a loan or meet contractual obligations. From an energy trading point of view, the term—"meet contractual obligations" gives rise to credit risk.

Mark-To-Market: Mark-To-Market is an accounting method used for risk management. Mark-To-Market can be understood as a method of measuring the fair value of accounts that can fluctuate over time, such as assets and liabilities.

Settlement risk: Settlement risk is the possibility that one or more parties will fail to deliver on the terms of a contract at the agreed-

upon time.

Operational Risk: the simplest definition is the uncertainties a company faces while carrying out daily operations/activities.

Political Risk: The risk faced by a trading organization due to political decisions, events or conditions that impact the profitability is called Political Risk.

Netting: Netting is a method of reducing risk by offsetting trade or a portfolio with a counterparty.

GAS, COAL AND ELECTRICITY TRADING

The other popular assets in energy trading are Electricity, Gas and Coal. All three assets have different lifecycles and hence different dynamics. For example, electricity is the fastest form in terms of settlement which can be in terms of minutes to gas and coal which have settlement stretched out over a long duration.

The following section of the chapter gives a high-level understanding of Electricity, Gas and Coal Trading. As the nature and dynamics are different, additional reading material needs to be sought for them.

5.1 Electricity

Electricity as a form of energy is different from oil, refined products or coal. Its dynamics are entirely different from financial trading. It is different because we cannot store electricity in the same form. Electricity can be stored when converted to a different form of energy. The other parameter it differs from is the way it is produced which is inflexible.

Moreover, now renewable has become part of the energy mix, which is unpredictable as it is dependent on wind or sunlight. The end consumer wants 100% availability. Lastly, the grid over which electricity is transmitted needs to be stable. In summary, the short-term market spot market becomes an important tool to balance an overall system.

Apart from producers and consumers of electricity, a couple of essential participants in the electricity trading markets ensure grid stability. The Independent System Operators (ISO) and Regional Transmission Organizations (RTO) are responsible for the grid balance between the producers and consumers. Another type of participant is Transmission System Operator (TSO, which is an operator responsible at the national or regional level entrusted to transport gas or electricity over a fixed infrastructure. TSO are present in both Electricity and Gas markets.

There are three ways to procure electricity. One is long-term contracts or power purchase agreements. The second is OTC contracts, and the third one is energy exchanges. Long-term contracts or power purchase agreements are contracts signed between producers and distributors. The opportunity for trading is available in OTC and energy exchanges. In the OTC market for electricity, trading is different as a physical settlement requires a grid for dispatch. The trading parties must interact with other participants to ensure grid stability. Since the deadlines for spot exchange deliveries are becoming shorter, OTC trading is becoming expensive. A long-term contract in the OTC market is the best way to minimize price risks for power producers. Moreover, exchanges are entering the electricity OTC market by offering standardized OTC contracts to reduce cost, enabling trading across all order types.

5.2 Gas

Natural Gas (NG) is considered cleaner than other fossil fuels. Hence it is gaining prominence in the fight against climate change. The

paradox is that one of the most significant components of natural gas is methane, and methane is one of the most potent greenhouse gases. Thus, natural gas when exposed to the environment is an issue, and hence escape of natural gas during extraction, transport, storage, and delivery needs to be minimized. Natural gas usage ranges from industrial to retail, making it a big market, hence making trading attractive.

Sources of natural gas

Following are some of the significant sources of natural gas.

1. Conventional Onshore – The gas extracted on land by a dedicated well generally uses a vertical drilling technique.
2. Associated – Natural gas extracted along with crude oil is called associated gas.
3. Conventional Offshore – Conventional dedicated gas wells drilled on water bodies using vertical gas techniques.
4. Tight Gas – Gas trapped and extracted from impermeable rock and non-porous sand formation.
5. Shale – Gas extracted from the shale formation. The techniques used for such extraction are either horizontal drilling or hydraulic fracturing (hydrofracking).
6. Coal Bed Methane – Natural gas recovered from coal seams through shallow horizontal drilling is called Coal Bed Methane.

Transportation of Natural Gas

Natural gas is transported differently on land and sea. Natural gas is predominantly transported overland via pipeline. In the case of seas, it

is done by ships. There are pipelines in the ocean that also transport natural gas.

Pipelines used to transport gas can be classified as Gathering Pipelines,

Interstate/intercountry/intercontinental pipelines and distribution pipelines. The Gathering Pipeline gathers gas from different wells. The interstate/intercountry/intercontinental transport from production to the consumption point. The distribution pipelines do the last-mile delivery to the end customer.

The natural gas transported to ships is transported as Liquefied Natural Gas (LNG). LNG vessels need to transport the same. Moreover, LNG terminals are required to load and unload the gas. LNG terminals do liquefaction to load it on the ship as the LNG vessel transports natural gas in liquid form. LNG terminals also do re-gasification after unloading to either feed to interstate/intercountry/intercontinental or distribution pipelines for further transportation.

Uses of Natural Gas

1. Industrial: The industry uses natural gas as a source of heat. Apart from being clean, the other reason to use natural gas as a heat source is the ease of use. Natural gas can be started and stopped on demand. In the case of coal, the coal needs to be burnt out completely. Besides being used a source of heat, natural gas is used for fertilizer, antifreeze, plastic, pharmaceuticals, chemicals and the fabric industry.
2. Domestic: Households use gas also as a source of heat. It is used for cooking, water heater, and heating boiler.

3. Power Generation: Natural gas being a clean fuel, has received interest from power generators. Power generators are increasingly using it in power plants. Due to high price fluctuations, some of the power plants run seasonally.

5.3 Coal

Coal is an essential source of energy. The above statement comes from coal-producing 37% of global electricity production. Since coal generates pollutants when combusted, countries are trying to reduce coal dependency. Though the shift in the energy mix is on the way, the demand will not go down anytime soon, as coal has many issues.

The primary use of coal is in electricity production. Similarly, coke from coal is a significant input to metal production. The steel industry is also a major user of coal. Coal is the primary source of energy for cement production. Since coal generates pollutants when combusted, countries are trying to reduce coal dependency. Though the shift in the energy mix is on the way, the demand will not go down anytime soon, as coal has many issues. Coal is the major source of energy for cement production. To use coal domestically is converted to gas and used for lighting, heating and cooking. Coal is on the way out but is used in the transport industry. Syngas from the gasification of coal can be further processed to produce blocks to build in the chemical industry like methanol, ammonia and urea. The advanced use of coal is to manufacture carbon fibre and special ingredients such as silicon metals.

The advantages of coal are it as accessible as it is available above ground. The accessibility is reduced with underground mines. The

usability of coal is high; Coal can be used directly after mining. Coal is used in multiple industries and making it a versatile ingredient. The abundance of coal in nature ensures it is an affordable source of income. Coal is safer to transport, store and handle than other fuel sources.

Coal can be classified into four main types. The four main types are (a) Anthracite, (b) Bituminous, (c) Sub-bituminous and (d) Lignite. The different types of Coal have different calorific values. Anthracite has the highest calorific of 30 millijoules per kilogram. Similarly, bituminous has 18.8-29.3 millijoules per kilogram. For sub-bituminous is 8.3 – 25 millijoules per kilogram. Lignite has the lowest range of calorific value (Brown Coal). It is 5.5 – 14.3 millijoules per kilogram. Coal is classified as Thermal Coal and Metallurgical Coal (“coking coal”). Metallurgical Coal (or “coking coal”) is mined to produce the carbon used in steelmaking, while thermal Coal is used to make steam that generates electricity.

The coal lifecycle can be categorized into (1) Mining, (2) Washing, (3) Transportation, and (4) Usage (electricity/heat). Mining is the process of extracting Coal from the ground. There are two ways of mining: surfing and underground mining. The washing of coal is done mainly to separate soil and rock from coal. In the process, the coal is crushed and segregated to store in stockpiles as per the grade of Coal. It must be transported from stocks at the mine head to the place of usage. Rail and waterways are used to transport coal. At the destination, coal is used as per the uses mentioned above. Most of the

use for electricity, gasification, steel plants and others are mentioned above.

Trading of coal is done by players categorized as importers and exporters. The unit of trading is 858.8 metric tons of oil equivalent. The major importers of coal are China, India, Japan and Europe. The major exporters are Russia, Australia and Indonesia. Twelve major importers and exporters dominate the coal trading market. Some major coal traders are Trafigura, Hind Energy and Coal Beneficiary India limited, Glencore, Mercuria and Vitol Holding B.V.

Coal trading will continue to dominate until alternate energy becomes cheaper for developing economies. The dynamics of oil trading mentioned at the start of the book applies to coal trading—the only thing to keep in mind is that few big players dominate it.

Chapter Summary

The following table summarizes some of the dynamics of different energy trading asset classes:

	Coal	Oil	Natural Gas	Electricity
Production	Overground or Underground mines.	Onshore and offshore oil wells.	Onshore and offshore oil wells.	Renewable and non-renewable generation units.
Transport	Rail or ship.	Tanker or pipeline.	Pipeline or LNG tanker.	Transmission cable.
Consumption	Need washeries.	Needs distillation.	Needs treatment.	Readily consumable.

Impact of Climate Change	Unpopular.	Treat electricity as a source of energy.	Increasing popularity.	Increasing popularity.
Demand	Decreasing.	Decreasing over the medium term.	Increasing.	Increasing.
Supply	Upward of 180 years depending on the type of coal.	Approximately 88 years based on recovery mechanism.	Would last for 165 years.	Available but unstable for renewable generation units.

Table: Comparison between different sources of energy

TECHNOLOGY IN ENERGY TRADING

In this section, we are going to discuss the current trends in technology. To depict a technology landscape for a trading desk is dependent on the organization. In general, a trading desk would have the following technologies:

6.1 Blockchain

The concept of blockchain has been floating around since 1982. Satoshi Nakamoto conceptualized the first blockchain in 2008. Blockchain was made famous for its usage in a cryptocurrency called Bitcoin in 2009.

Blockchain technology employs proof-of-work to record a public history of the transaction, and the consensus mechanism can enforce any needed rules and incentives. The four key elements of Blockchain technology are (a) Transparency, (b) Trust, (c) Efficiency and (d) Control and security. There are four types of blockchain networks – public blockchains, private blockchains, consortium blockchains and hybrid blockchains.

A public blockchain is non-restrictive and permissionless, and anyone with internet access can sign on to a blockchain platform to become an authorized node. A private blockchain works in a restrictive environment like a closed network or is controlled by a single entity. A type of blockchain that tries to be used best of both public and private elements of blockchain is called a hybrid blockchain. Blockchain technology is similar to hybrid technology and is known as

a federated blockchain, but multiple organizations collaborate on a decentralized network. A consortium blockchain is a private blockchain with limited access to a particular group, eliminating the risk that comes from one entity controlling the network of a private blockchain.

In the case of the oil and gas industry, multiple consortia exist and trying to implement a decentralized network for their members. Some consortiums and companies offer solutions related to oil and gas companies. Some of the use cases are focused on energy trading and risk management. VAKT is one such consortium. Some investors are BP, Shell, Saudi Aramco, Chevron, TotalEnergies, Reliance Industries, Equinor, Gunvor, KOCH Industries Inc, Mercuria, ABN Amro, ING and Societe General. Some of the solutions offered are Deal Recap and Confirmation, Trade Finance, Logistics and Invoice data. Another similar blockchain company is Komgo. It is owned by AMN Amro, BNP Paribas, Citi, Consensusys, Credit Agricole, Credit Suisse, Gunvor, ING, KOCH, Macquari, Mercuria, MUFG, NATIXIS, Rabobank, SGS, Shell, SMBC, Societe General and Total Energies. Komgo offers solutions in streamlining trade finance, market closing the deal, know-your-customer and document tracking. Another company at the time of writing is we trade. The company calls itself the world's first enterprise-grade blockchain-enabled trade finance platform. Its investors are mostly banks (around 12). Since trade finance is an essential element in energy trading, this company also becomes a blockchain solution providing a company to consider blockchain-enabled solutions. There are similar many consortiums and companies offering blockchain solutions.

Some of the uses cases that blockchains are trying to solve can be summarised as follows:

Smart Contracts: We know that contracts are lengthy and complex from the trade life cycle. They sometimes span multiple geographies and timeframes. Smart contracts can help streamline the trade finance process, which is paper-based and consumes lots of time. A smart contract would record a deal in technical language and not a legal language, creating a peer-to-peer network. The smart contracts would become self-verifying pieces of code and execute the agreements when conditions for trade logic are met.

Compliance: Data shared over a blockchain is transparent, and it could help in compliance. One such example is that shared counterparty data will enhance regulatory compliance. To explain that if that particular counterparty is blocked in geography, all the participants of the Block Chain will update their systems. Some regulatory frameworks like Dodd-Frank and European union directives on transparency increased transparency for data stored. The data is shared with regulators in real-time, increasing regulatory and internal compliance.

Product Tracking: The crude oil extracted has unique characteristics. Thus the supply of quality can differ from contracted crude. Such a scenario can lead to disputes. Solving such disputes can take time and impact the working capital requirements of the companies. Moreover, there are chances of oil flow from sanctioned countries and companies. So, tracking of products becomes an essential parameter for oil companies. Blockchain offers a tamper-proof distributed ledger

in real time. Thus blockchain can solve the problem of tracking crude oil through the entire supply chain and bring in operational efficiencies.

Cybersecurity: Many communications in energy trading happens on unsecured networks—for example, a chat over a communicator over the internet, a phone call or an email. Blockchain is secure and stores data over a distributed network. Along with security in storage, it brings the integrity of the data stored. A tamper-proof communication, storage, and data transmission can increase security in the oil and gas industry, including energy trading.

There are many more areas in energy trading, especially renewable energy bills and electricity trading. Also, there are opportunities to open peer-to-peer markets using blockchain. The technology and use cases are evolving for blockchain and are slowly but indeed getting implemented.

6.2 Robots Process Automation

Robotics Process Automation (RPA) is a non-invasive rule-based software technology that automates repetitive tasks and processes which humans previously did. RPA is what makes the process robust by eliminating human error. Moreover, the bots are available 24x7 eliminating the risk of unavailability.

As per the trade life cycle, we know many teams are required to make a trade successful. One such team is the pricing team. The pricing team ensures all the prices are correctly entered into the system. The team has to verify the data against many sources. Moreover,

monitoring the prices of the competition is an important activity. All the processes are manual hence robots can help to automate those.

Various reports need to get generated. A self-reporting tool can solve the problems of customized reporting. There are standard reports required for regulatory reporting, and reports to mark margin calls. Bots can get involved and automate the process, bringing in efficiency and reducing the risk.

Many stakeholders present invoices during the entire trade life cycle. The invoice process is document-intensive but standard and rule-based. The same can be automated using bots.

Based on the trade status, we can use bots for scheduling, nomination, and dispatch. Depending on the digitization adoption of the trading organization, either part or the entire process can be automated.

Gas can be in the pipeline, storage assets, or contracts and yet to be delivered or in the voyage. We can program the bot to interact with multiple systems to capture data and create a view. With the help of analytics and insights, the bots can be more intelligent and not just follow predefined rules.

The other area where RPA is making a considerable impact is automating the meter reading and metering process, including meter management, customer billing, account management, consumption management, segmentation and exception processing. The same robots with analytics and insights can provide real-time information to traders for demand.

Above are a few examples or use cases where RPA would transform energy trading. Implementation of RPA is a cost, and there is a

question about calculating the return on investment. There are multiple ways to calculate the return on investment. The first one is a simple calculation of full-time employee work being automated. The second one is a reduction in errors/exceptions from the past. The third would be the team handling more volume and value of work than before. The most crucial benefit of RPA is the movement toward integrating AI to make an intelligent trading ecosystem.

6.3 Data, Analytics and Insights

What is data?

Data is the information obtained throughout the trade life cycle for energy trading. During the entire energy cycle, terabytes of data are collected. Even if considering the trading area, there is a massive amount of data that is generated both internally and externally. The following table shows some of the areas of data generation:

External Data Sources	Internal Data Sources
Price Information (OTC Market Data)	Generation volume (If integrated company)
Exchange information	Deal Volume Trades
Weather information	Contract
Public events	Forward curves (Calculated internally)
Market trading volume	Trading limits Risk
Market generation and consumption reports	Operations

Table: Sources of data in Energy Trading

Some of the data is structured data, and some are unstructured. Most of the information is on real-time bases, so capturing the exact needs to be the focus. But without historical data, the use of live data is limited, and hence storage is also paramount. Hence for trading firms capturing, storing, and availability of data for usage has to be the first step of the strategy.

What is Analytics?

For a layperson, analytics is the discovery of patterns and trends from the data. For energy trading, one of the simple trends can be what is the impact of seasonality (high-level changes in the prices based on the month of trading)

What is Insight?

Insight is the value obtained using analytics. For energy trading, some of the simple use cases for insights are:

Process Optimization – Trading is a complex process; the insights gained based on patterns of trade fulfilment can be used for process optimization. Process optimization can be in terms of people, automation or even in times of operational risk reduction.

Trading Risk – There can be patterns in that a counterparty always makes a late payment or a dispute with the counterparty for the quality delivered or operations issues. Then the system can alert the trader when the trader is trying to trade with the counterparty. Based on the alert, the trader can take mitigation actions to reduce the company's overall risk.

With correct data and analytics, strategy can be a competitive edge for the trading desk.

6.4 Machine Learning (ML) and Artificial Intelligence (AI)

Any technique enables machines to mimic human intelligence by collecting and processing data. The method that allows machines to improve their intelligence as they are exposed to more data over time is Machine Learning. The definitions clearly show that both terms are different, but they are used interchangeably in the industry.

Though algorithmic trading is increasing in oil trading, the best use case for AI and ML is for electricity trading. The complexity of intra-day electricity trading is the highest; hence, a machine is needed to help the trader make informed decisions.

Though energy trading is behind the learning curve in technology adoption, one of the primary use-case the trading firms use is AI and ML in forecasting. Demand forecasting is one of the basic types of forecasting done with AI's help. The same is observed across all types of energy. The most return on investment comes in electricity trading. Automated trading based on AI and ML is also increasing. The contribution of algorithmic trading as a percentage of total trading on commodities exchanges has increased substantially.

Chapter Summary

Information arbitrage and information processing are competitive advantages for a trading desk. Trading desks require investment in technology to retain a competitive edge. Artificial intelligence,

machine learning, Data and Analytics are some technology levers that trading companies use to ensure competitive advantage.

Risk and risk mitigation is one of the major for a trading desk. Technology like Block Chain, Artificial Intelligence, Analytics & Insights is some of the technologies that trading desks rely on to manage risk effectively.

Efficiency and optimization directly reduce cost per trade. Digitization, Robotics Process Automation, Analytics & Insights are some of the technologies being explored by trading desks to achieve the same.

BACK TO BASICS

7.1 Understanding Crude Grades and Benchmark

American Petroleum Institute measures the density of crude in terms of American Petroleum Institute Gravity (API gravity). The number ranges from 10 – 70. For a common understanding, we know that oil floats on water. But it is the lighter crude oil (means whose gravity number is higher) that floats on water. The heavy crude oil (whose gravity is lower) will sink into the water.

From a usage point of view, the crude with lower density (lighter the crude oil – higher the API number) is easy to refine into gasoline, diesel, and other distillates. Heavy crude oil is difficult to refine. Meaning a lighter crude would give more amount of petrol or diesel fuel. Since gasoline and diesel fuel are priced the same, they will sell the lighter crude at a premium to the heavier crude. Also, remember that heavier crude needs an advanced methodology to process and refine.

One of the ways to look at crude oil is by looking at the quality of the crude oil. The quality of the crude oil means the sulphur (sulfur) content of the crude oil. Lower sulphur content crude oil is called “Sweet”. Similarly, the higher content crude oil is called “Sour”. The benchmark to define is 0.5%. If the Sulphur content is less than 0.5%, it is called “Sweet”. Anything higher than 0.5% is called “Sour”.

The Sweet is easier to refine than the Sour. Similarly, to light crude, sweet crude commands a premium compared to sour crude. Just like

lighter crude, sweet crude is easier to refine. So sweet and light crude oil would trade at a premium to heavy and sour crude.

Between light and heavy, sweet, and sour we can establish the different types of crudes. The following chart shows different grades of crude oil.

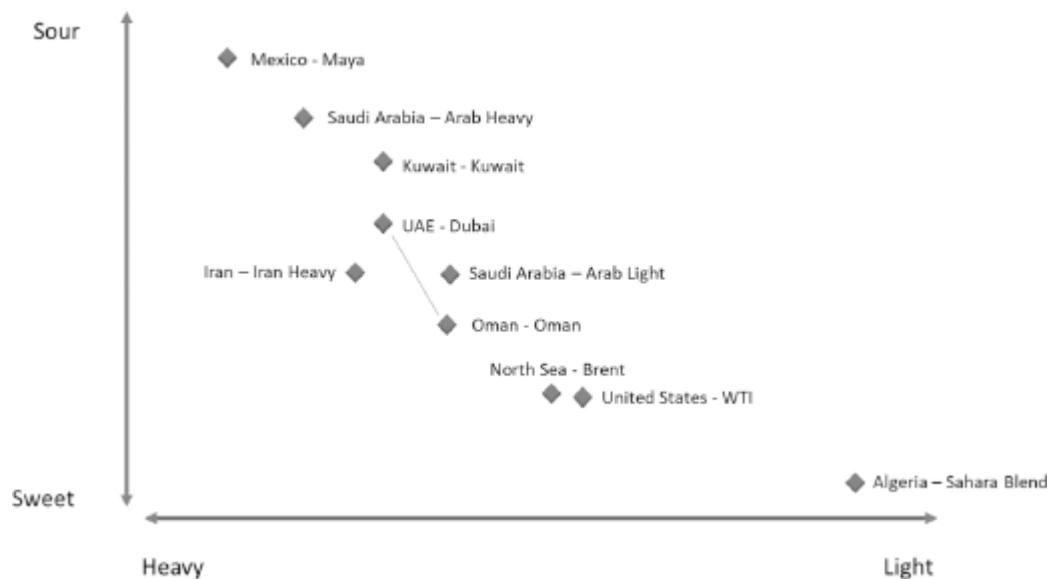


Figure: Different grades of crude classification

Since all the oil is extracted across different geographies worldwide, they all have different densities and sulphur content. Thus, making each crude oil that is drilled, different. To identify the type of crude, oil companies follow a process called “Crude Oil Assay”. An assay is a process of chemical evaluation of feedstocks by testing laboratories.

To price different types of crude oil, there is a need for a benchmark. The famous oil grade is Brent, Western Texas Intermediate (WTI) and Dubai/Oman. The other grades of crude oil will trade at a premium or discount to these benchmarks.

Brent: Brent grade of crude oil is extracted from the oil fields in the North Sea. The field names are Brent, Forties, Osberg, or Ekofisk.

Brent crude is considered light and sweet. The transportation is done mostly in crude oil tankers. It is estimated that more than 60% of oil traded is traded regarding Brent Crude.

Western Texas Intermediate (WTI) is extracted from oil fields in the USA. The crude oil is extracted generally from the oil fields of Louisiana, North Dakota, and Texas. WTI is considered very sweet and very light. The fields are landlocked, and hence the transport is through trains and pipelines. The benchmark is used to price crude oil produced in Gulf and Mexico, Canada, and South America.

Dubai/Oman grades of crude are medium and sour. The Dubai/Oman grade oil is used to price the crude. The crude grade is considered medium and sour. One of the primary users of this benchmark is the Saudi Oil company Aramco for benchmarking the oil it sells to Asian clients.

Three crude grades Brent, WTI and Dubai/Oman are the major benchmarks for the oil trading that happens worldwide.

7.2 Difference between Energy Trading and Equity Trading

Capital Markets traditionally had four different asset classes (a) Equity, (b) Debt, (c) Forex and (d) Commodities. The independent trading companies, commodities producers and consumers were active players. Regulation in commodities trading ensured that commodities trading was no more attractive to banks for proprietary trading. Thus, commodities trading is left to independent commodities trading firms (big and small) and producers and consumers of commodities. The following table demonstrates the difference between Commodities and Equity Trading.

Characteristics	Commodities Trading	Equity Trading
Nature of the product	Commodities are raw materials or products that are used to manufacture finished goods or are directly consumed.	Equities are financial instruments that represent ownership in a company.
Quality	Commodities or products have several varieties or grades, and each lot can differ from one another.	One security unit does not differ from another of the same type in terms of its face value and characteristics.
Supply	The supply of a commodity is not fixed.	The number of equity of a company is fixed.
Purpose	Used for hedging against price fluctuations or benefit from price fluctuations.	Equity purpose is to build wealth through capital appreciation and dividend income. Trading help generates profit from price fluctuations.
Volatility	Commodities prices are highly volatile.	Relatively equities are less volatile as an asset class.

Bid/Ask Spread	Can be high for most commodities.	It is done in a highly competitive environment and hence minimal.
Trading Horizon	The trading horizon is generally short-term. Most of the trading happens in the Futures market.	It is suitable for both the short-term and the long-term. The trading duration varies from intraday, day, month, year or even many years.
Trading Hours	24x7	Limited to geographics laws in which they are traded.
Regulation	Trading is lightly regulated.	Equity trading is done in a highly regulated market.

Table: Difference between Commodities Trading and Equity Trading

7.3 Incoterms: Making international trade Simpler

One ship blocked the Suez Canal in March 2021 for more than a week. The delay caused due to this stoppage of ships brings fore the question, bearer of the risk and costs of the good stuck in transit. A simple question like, “who would make an insurance claim?” would have kept lawyers busy for months to come. It would have been a nightmare if there were no clearly defined internationally accepted

terms of the contract. The reduction in complexity is due to the usage of Incoterms by the International Chambers of Commerce.

The International Chamber of Commerce was founded in 1919 by a handful of entrepreneurs representing the business in general as a private organisation. The mission of ICC is “We make business work for everyone, every day, everywhere”. To facilitate international trade in 1933, the organisation publishes the first rules for uniform customs and practice for documentary credits (UCP).

In 1936, the first edition of the Incoterms rules was published featuring six trade terms. Incoterms is an acronym for international commercial terms. Incoterms rules are widely used in international commercial transactions or procurement processes, and their use is encouraged by trade councils, courts and international lawyers. A series of three-letter trade terms related to common contractual sales practices, the Incoterms rules are intended primarily to communicate the tasks, costs, and risks associated with the global or international transportation and delivery of goods. Incoterms inform sales contracts defining respective obligations, costs, and risks involved in delivering goods from the seller to the buyer. Still, they do not conclude a contract, determine the price payable, currency or credit terms, govern contract law or define where title to goods transfers.

Since the first publication of Incoterms in 1936, they have evolved and revised in 1953, 1967, 1976, 1980, 1990, 2000, and 2010. The ninth version, Incoterms 2020 were published in September 2019.

The Incoterms help define the following at a high level:

- (a) Obligations: Who does what as between seller and buyer. For example, who organises carriage or insurance of the goods or who obtains shipping documents and export or import licences.
- (b) Risk: Where and when the seller “delivers” the goods, in other words, where risk transfers from the seller to the buyer.
- (c) Costs: Which party is responsible for which costs, for example, transport, packaging, loading, or unloading costs, and checking or security-related costs?

The Incoterms rules do NOT deal with the following matters:

- whether there is a contract of sale at all
- the specifications of the goods sold
- the time, place, method or currency of payment of the price
- the remedies for breach of the contract of sale
- most consequences of delay and other violations in the performance of contractual obligations
- the effect of sanctions
- the imposition of tariffs
- export or import prohibitions
- force majeure or hardship
- intellectual property rights or
- the method, venue, or law of dispute resolution in case of such breach.

Perhaps most importantly, the Incoterms rules do not deal with the transfer of property or title or ownership of the goods sold.

Eleven terms are defined in Incoterms 2020. Seven of the 11 terms are defined for any mode of transport, and four for Sea transport or Inland Water Way transport. The way to use incoterm is “Term: < named place of delivery >”. For example, EXW < Company X factory in London >.

A summary overview of the Incoterms is as follows:

EXW – Ex-Works or Ex-Warehouse

The seller places the goods at the buyers’ disposal at the seller’s premises or any other named place. For example, the place can be works, factory, or warehouse. The buyer has to arrange the loading of goods on the collecting vehicle. The buyer needs to clear them for export where such clearance is required.

FCA – Free Carrier

The seller delivers the goods to the carrier or another person nominated by the buyer at the seller’s premises or another named place. The buyer and seller are advised to specify the point at the delivery place where the parties’ risk passes.

FAS – Free Alongside Ship

The seller is responsible for delivering the goods alongside the vessel on a quay or barge at the named port of shipment nominated by the buyer. The risk of loss or damage to the goods passes when the products are alongside the ship. The buyer bears all costs from that moment onwards.

FOB – Free On Board

The seller delivers the goods on board the vessel nominated by the buyer at the named port. The risk of loss or damage to the goods passes when the products are on board the vessel. The buyer bears all costs from that moment onwards.

CFR – Cost and Freight

The seller delivers the goods on board the vessel. The risk of loss or damage to the goods passes when the products are on board the vessel. The seller must pay the cost and freight to the shipping company to bring the goods to the named port by the buyer.

CIF – Cost, Insurance and Freight

The seller delivers the goods on board the vessel. The risk of loss or damage to the goods passes when the products are on the ship. The seller must pay the cost and freight to the shipping company to bring the goods to the named port by the buyer. The seller must also pay for the insurance for the entire journey until the buyer's named port.

The buyer should note that the seller must obtain insurance only on the minimum cover under CIF. Should the buyer wish to have more insurance protection, it will need to either agree as much expressly with the seller or make its insurance arrangements.

CPT – Carriage Paid To

The seller delivers the goods to the carrier or another person nominated by the seller at an agreed place (if any such site is agreed upon between parties). The seller must contract for and pay the carriage costs necessary to bring the goods to the named place of destination.

CIP – Carriage and Insurance Paid To

The seller has the same responsibilities as CPT, but they also buy insurance for the goods during the journey. The buyer should note that the seller must obtain insurance only on the minimum cover under CIP. Should the buyer wish to have more insurance protection, it will need to either agree as much expressly with the seller or make its insurance arrangements.

DAP – Delivered At Place

The seller delivers when the goods are placed at the disposal of the buyer on the arriving means of transport ready for unloading at the named place of destination. The seller bears all risks involved in bringing the goods to the named place.

DPU – Delivered At Place Unloaded

The seller delivers when the goods, once unloaded, are placed at the disposal of the buyer at a named place of destination. The seller bears all risks involved in bringing the goods to and unloading them at the named place of destination.

DDP – Delivered Duty Paid

The seller delivers the goods when the goods are placed at the disposal of the buyer, cleared for import on the arriving means of transport ready for unloading at the named place of destination. The seller bears all the costs and risks involved in bringing the goods to the place of destination. They must clear the products for export and import, pay any duty for both export and import and carry out all customs formalities.

The following chart can be used as a ready reference for the Incoterms.

	Freight Collect Terms						Freight Prepaid Terms				
Groups	Any mode or modes of Transport		Sea and Inland Waterway Transport				Any mode or modes of Transport				
Incoterm	EXW	FCA	FAS	FOB	CFR	CIF	CPT	CIP	DAP	DPU	DDP
Transfer of Risk	At Buyer's disposal	On Buyer's transport	Along side ship	On Board Vessel	On Board Vessel	On Board Vessel	At Carrier	At Carrier	At Named Place	At Named Place unloaded	At named place
Export Packing	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Loading Charges	Buyer	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Delivery to Port/ Place	Buyer	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Export duty, Taxes & Customs clearance	Buyer	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Origin Terminal Charges	Buyer	Buyer	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Loading on Carrier	Buyer	Buyer	Buyer	Seller	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Carriage Charges	Buyer	Buyer	Buyer	Buyer	Seller	Seller	Seller	Seller	Seller	Seller	Seller
Insurance	Negotiable	Negotiable	Negotiable	Negotiable	Negotiable	Seller	Negotiable	Seller	Negotiable	Negotiable	Negotiable
Destination Terminal Charges	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Seller	Seller	Seller	Seller	Seller
Delivery to Destination	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Seller	Seller	Seller
Unloading at Destination	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Seller	Buyer
Import duty, Taxes & Custom clearance	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Seller

Figure: Risk owner as per Incoterm

Please note Incoterms is a registered trademark of the International Chambers of commerce.

For further information on Incoterms please visit <https://iccwbo.org/>

7.4 Letter of Credit

Two critical documents come up when discussing “International Trade” or “Cross border sales”. The documents are (a) a Letter of Credit and (b) a Bill of Lading. A Bill of Lading is a document issued by a carrier (someone who transports and delivers goods) to the shipper (someone who supplies and/or owns goods), confirming the goods were received in acceptable condition and ready to be shipped.

The importer brings goods from overseas or across the border to be consumed/sold in their own country. The counterparty supplying

goods needs a guarantee that the counterparty will get paid once the goods are delivered. A letter of credit (or L/C as known in the parlance of international trade) is a document issued by the banks of the importer that guarantees the payment.

A letter of credit (LC) is defined as a document issued by an importers bank that promises a payment after certain conditions are met. The LC is also known as “Documentary Credit”, “Bankers commercial credit” or “Letter of Undertaking”.

There are four major participants involved in the Letter of Credit. The participants are (a) Applicant (Buyer/Importer) (b) Issuing Bank (Buyer/Importer Bank), (c) Advising Bank (Negotiating Bank, Exporter’s Bank) and (d) Beneficiary (Exporter/Seller/Counterparty). The following diagram depicts an LC process:

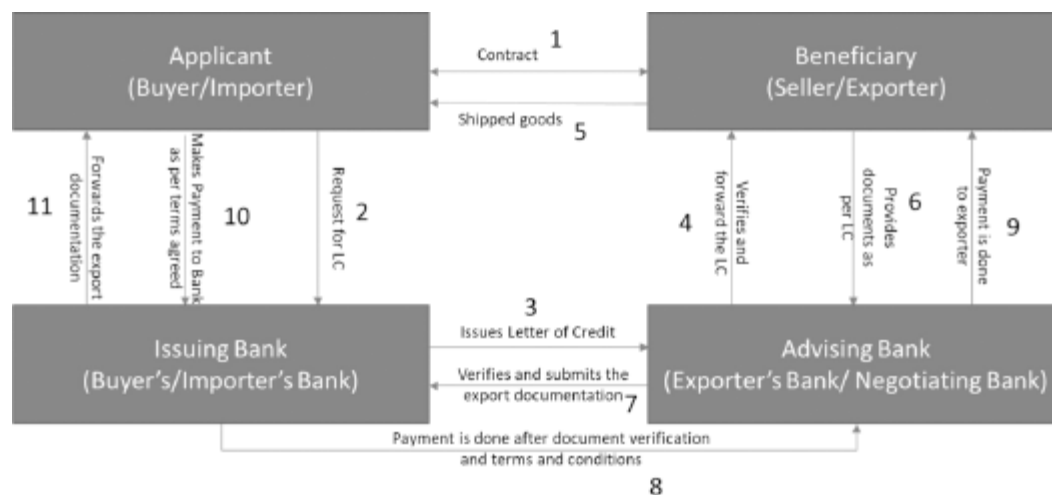


Figure: Typical Letter of Credit process

The following 11 steps describe the entire LC process.

Step 1: The importer and sellers agree to a contract. The mode of payment agreed upon is a letter of credit.

Step 2: The importer approaches a bank (Issuing Bank) to issue a letter of credit to the counterparty/seller/exporter bank with the terms and conditions of the contract. One of the main conditions is the documents that need to be presented by the exporter verifying the goods have been shipped (the Original Bill of Lading is one of the documents)

Step 3: The issuing bank issues the letter of credit with the terms and conditions as instructed by the importer to the sellers/exporters/counterparty's bank (Advising bank/Exporter's Bank/Negotiating Bank).

Step 4: The Advising bank (Exporter's Bank) verifies the Letter of Credit and sends it to the seller/exporter/counterparty.

Step 5: The seller/exporter/counterparty ships the goods to the buyer/importer. The seller collects the export documentation.

Step 6: The seller/exporter/counterparty submits the export documentation to the Advising Bank (Exporter's Bank)

Step 7: The Advising Bank verifies the export documentation as per the LC terms and conditions and submits it to the Issuing Bank.

Step 8: The Issuing Bank verifies the export documentation and, pays the Advising Bank once satisfied.

Step 9: The Advising Bank makes payment to the seller/exporter/counterparty upon receipt of the payment.

Step 10: The issuing bank informs the importer/buyer of the receipt of documents and requests payment. The importer/buyer makes the payment to the issuing bank.

Step 11: Upon receipt of the payment, the issuing bank releases the export documentation to the buyer/importer. The importer takes the documentation and gets the goods shipped by the seller/exporter/counterparty.

Thus, a letter of credit ensures that the default risk is effectively mitigated in an international trade where the importer/buyer and the seller/exporter/counterparty might not know each other.

The advantages of having a letter of credit as a method are listed as follows:

- (a) The seller/exporter has the confidence to receive payment upon shipping the goods. The seller/exporter would receive payment even if the buyer/importer enters bankruptcy after shipping goods.
- (b) The delivery of the shipped goods is not required to receive the payment. For example, when the Suez Canal was blocked, if the seller/exporter/counterparty had shipped the goods and submitted the documentation, payment would be received if the issuing bank was satisfied.
- (c) For the buyer, the banks verify the export documentation to ensure the goods are shipped as per the contract and hence mitigate the risk of fraudulent invoices from the seller/exporter/counterparty.

There are different types of letters of credit. Some of the common ones are described as follows:

Stand by Letter of Credit (SBLC): SBLC is a letter of credit then does not make a payment unless there is proof of the buyer/importer

defaulting on the payment. An SBLC by design is not to make a payment.

At Sight: The issuing bank makes payments to the advising bank when the documents are received and verified.

Revolving Letter of Credit: A typical LC would cover only one transaction. A revolving letter of credit would cover multiple transactions and be drawn out for a longer duration of time.

Open Letter of Credit: A typical LC cannot be discounted, but an Open Letter of Credit recipient can get it financed.

Back-to-Back Letter of Credit: If there is an intermediary, for example, a broker, then a Back-to-back letter of credit is used for the transaction. There are two independent LC transactions in Back-to-Back LC. The buyer/importer would issue a letter of credit to the intermediary bank. The intermediary's bank would issue a letter of credit to the sellers/exporters/counterparty bank.

INTERESTING STORIES

8.1 Metallgesellschaft – An idea gone terribly wrong

Metallgesellschaft was incorporated in 1881 in Germany. As per the article of association, their main business was trading and manufacturing of metals and metal oxides. The company grew into one of the biggest German conglomerates. In the early 1990s, it decided to open an energy trading office in the US. The aim of the trading office was (a) to sell refined products in the Forward market, and the physical market, (b) invests in refining capacity to produce the products and (c) Hedge Forward sales using derivatives.

The strategy was simple and might be used by hedgers. In 1992, current physical prices were lower than the Futures prices. Thus, the sales contracts were set at higher prices in the future. The purchase of a near-monthly contract was cheaper as the prices were lower. The strategy was to cover the sales of the far month by purchasing the near-month contract and rolling over on maturity. The strategy would create profit if the market kept the way it is.

The one problem with the strategy was cash flow management. As the payment of a long-term sales contract would be realised over the long term, if the prices increase in the near term, payment needs to be made for margin calls on the short-term contracts and more cash is required for rolling over the contracts.

In 1993, the same thing happened. The prices in the short term started rising. The market reversed from “Backwardated” to “Contango”, and

management closed all the short-term positions. The closing of the contracts resulted in a loss of billions. Over a period in 1993, Metallgesellschaft made a loss of \$1.5 Bn to \$ 2.2 Bn. The group had to ask for a bailout, and the conglomerate had to restructure. It has been debated that the group made a hasty decision to close the contracts otherwise was a sound trading strategy.

The above incident at Metallgesellschaft showed that energy trading is not for the faint-hearted and is fraught with a loss without risk management or cash management. Moreover, oil trading is multi-dimensional and can quickly fluctuate between Backwardation and Contango.

8.2 Oil trading changed forever – COVID onset bets

Hin Leong Trading was founded in 1963 by Lim Oon Kuin and was headquartered in Singapore. Hin Leong was one of the largest independent commodities traders in Singapore. Fast forward to June 24, 2021, Singapore files 105 charges against the founder of Hin Leong. The charges comprise 25 forgery-related charges, 68 counts of cheating, 36 of conspiracy to commit forgery and one of conspiracy to forge valuable security (Bill of Lading). Hin Leong filed for bankruptcy protection in April 2020.

As many as 23 banks exposed \$3.5 Bn to Hin Leong Trading debt. It is estimated that HSBC alone has an exposure of more than \$ 600 Mn, with DBS \$290 Mn and OCBC \$ 200 Mn.

Hin Leong Trading going bust is an interesting case of the market view of a trading firm going wrong. On Dec 31, 2019, China informed WHO of pneumonia cases in Wuhan City without unknown cause.

Global oil prices started going down. Lim Oon Kuin in Singapore assumed that China would successfully control the virus. He believed that the spread would be an epidemic and not a pandemic. Thus, they started buying vast stockpiles of fuel.

The reverse happens; China goes into lockdown, and then the world. The oil prices plummeted nearly 70%, resulting in huge losses, including a banking scandal and banks were left exposed to Hin Leong Trading.

8.3 The day future prices went negative – Oil oil, everywhere oil and not a tank to store

The day was Monday, 20 April 2020, made history in the world of oil. It was the day the US oil prices turned negative. History was made due to one of the biggest drops in demand for oil in 25 years. The flood of unwanted oil in the US caused Western Texas Intermediate (WTI), a benchmark Oil price, to plummet to almost – \$40 a barrel. That meant US Oil producers were paying buyers to take oil off their hands.

Oil prices had been steadily declining since the coronavirus outbreak first broke out in China at the end of 2019. Almost all countries worldwide went into lockdown to curb the spread of the virus. That resulted in the demand for oil for transport nearly wiped out. The oil producers continued to pump oil, resulting in a supply and demand imbalance.

As the oil producers were pumping oil, the storage facilities worldwide were getting filled up to the brim, including in the US. The problem was acute in the US as most of the production was consumed there. Thus, once the market realised that the storage capacity was

nearly full, the price for the May contract went negative. Other benchmarks did not go negative, but Brent Futures were trading at \$20 a barrel.

One must understand that these are future prices; this does not include the cost of transport and refining or any other cost, so no one got paid to fill their cars for free or got money to fill their vehicles. The oil fields shut down, and once the future contracts for June started getting traded, the prices stabilised. The negative prices were short-lived. From Tuesday, 21 April 2021, trading started and fetched higher prices.

Turning of the price to negative, though short-lived, proved that oil pricing and eventually trading is multi-dimensional.

DEFINITIONS (A-Z)

Assay: A detailed chemical analysis of crude oil to predict the volume and quantity of refined product from refining the crude oil in the refinery. For a trader, a crude oil assay helps them find a good market for their crude. It can also help them find the right place to source the crude.

Backwardation: Backwardation happens when the future contract price is lower than the underlying price in the spot market. For example, the oil price for future contracts to be delivered after one month is \$100 per barrel. The price of oil in the spot market is \$90 per barrel. The market, in this case, is called to be in Backwardation.

Crack Spread: The simultaneous purchase and sale of crude oil and the refined product is called Crack Spread.

Contango: Contango happens when the price of a Futures contract is higher than the price of the underlying in the spot market. For example, the spot price of crude is \$80 per barrel, but the future contract price to be delivered after one month is \$90 per barrel.

Default Risk: The risk of the borrower or purchaser not being able to make the necessary payment is called Default Risk.

Enhanced Oil Recovery (EOR): A recovery process to recover oil after the primary and secondary oil recovery techniques are implemented.

Fractional Distillation Unit: A unit in the refinery that uses the Fractional Distillation process that separates the oil into different products. It is the standard delivery point for the NYMEX.

Gallon: The unit of measure in the oil industry like a Barrel. There are 42 gallons in a barrel. Approximately 3.78541 litres in a US gallon.

Henry Hub: The distribution hub in Erath, Louisiana for interstate and intrastate pipelines. The importance of the hub is that it is named as pricing point of natural gas.

IPE: The International Petroleum Exchange in London.

ICE: Intercontinental Exchange is an American company owning exchanges, clearing houses, technology, data and listing services.

Jarrow Turnbull Model: One of the first models to price credit risks.

KYC: It is an abbreviation for Know Your Counterparty.

Load Management: It is the process of managing the supply of electricity into the network.

Mark-To-Market: A fair value accounting method of asset or liability based on the current market price.

NGL: NGL is an abbreviation of Natural Gas Liquids. Natural Gas Liquids include Ethane, Propane, Butane, Isobutane and Natural Gas Condensate.

OPEC and OPEC Basket: OPEC is the Organization of the Petroleum Exporting Countries. It is a group of the 13 largest oil-exporting countries. OPEC aims to try and set the oil prices. OPEC Basket is the weighted average price of all grades sold by 13 member countries.

Primary Oil Recovery (POR): A process that uses the difference in pressure between the surface and the underground reservoir to recover oil is called the Primary Oil Recovery method.

Quota: OPEC sets a quota for each of the 13 producing members.

Revolving Credit: Revolving credit is an agreement between a trader company that allows a counterparty to trade up to a dollar limit while paying only part of the trade to the limit.

Spread Trading: Buying one commodity and selling another commodity to profit from the price gap.

Tanker: Vessels to transport crude and other commodities in the trade. Some examples are VLCC Very Large Crude Carrier, ULCC Ultra Large Crude Carrier, Panamax, Aframax, Suzemax are some of the types of tankers.

Unsystematic Risk: A risk that is unique to a specific company or industry.

Value at Risk (VaR): The way to measure the market risk is to use a statistical method called Value at Risk (VaR).

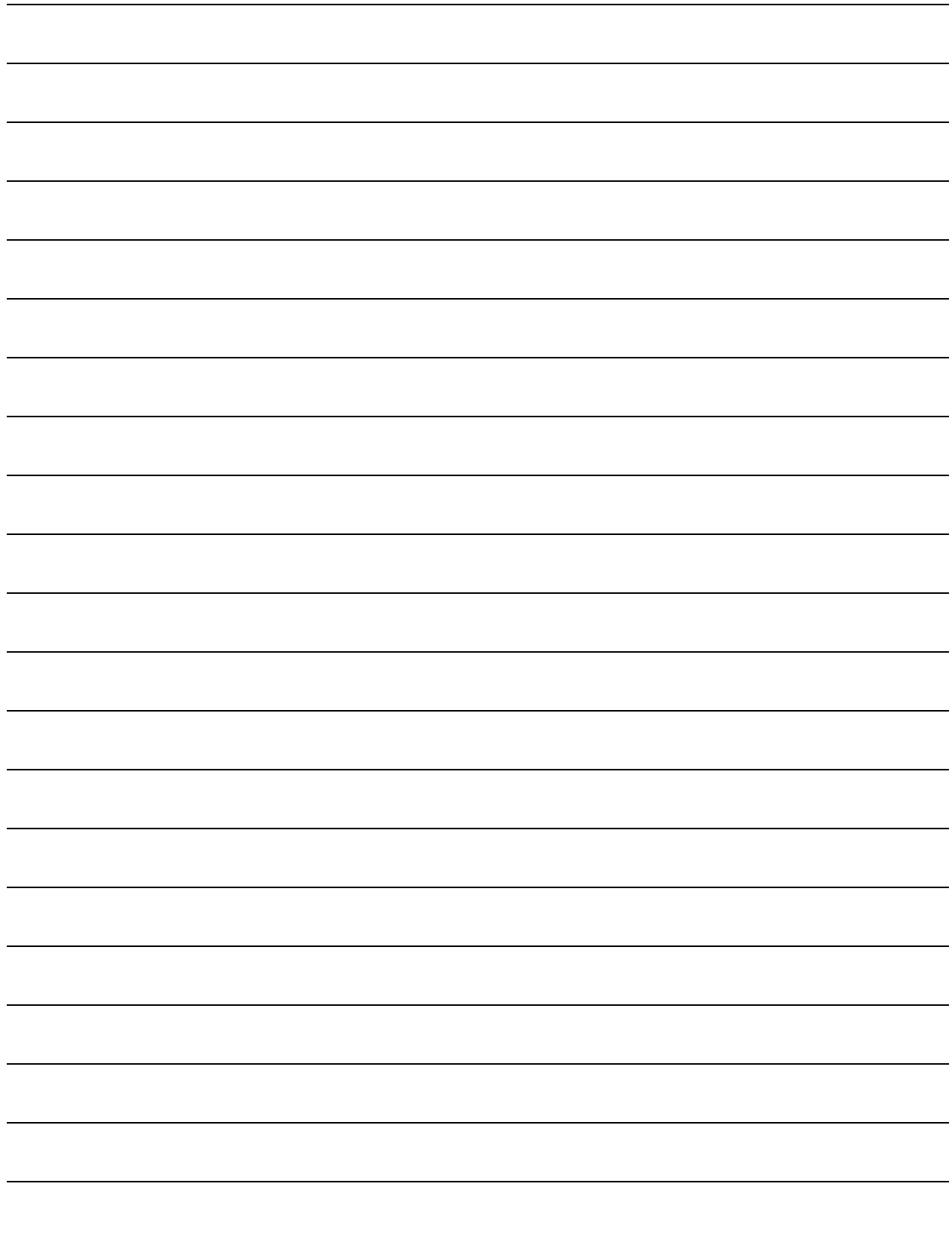
WTI: WTI stands for Western Texas Intermediate. It is a light, low-sulphur (“sweet”) grade of crude oil from the United States.

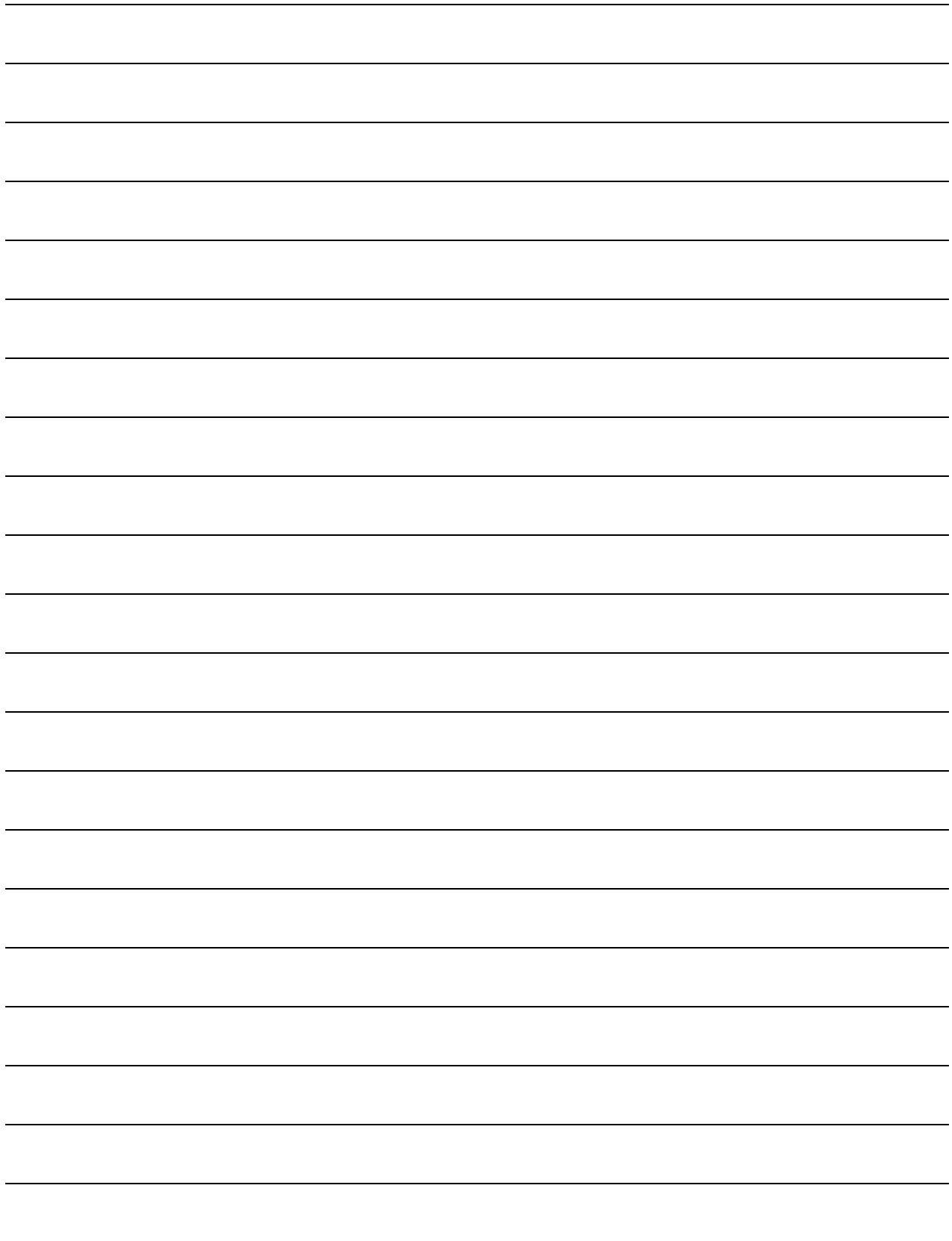
Yo-yo: Yo-yo is a slang term for a very volatile market.

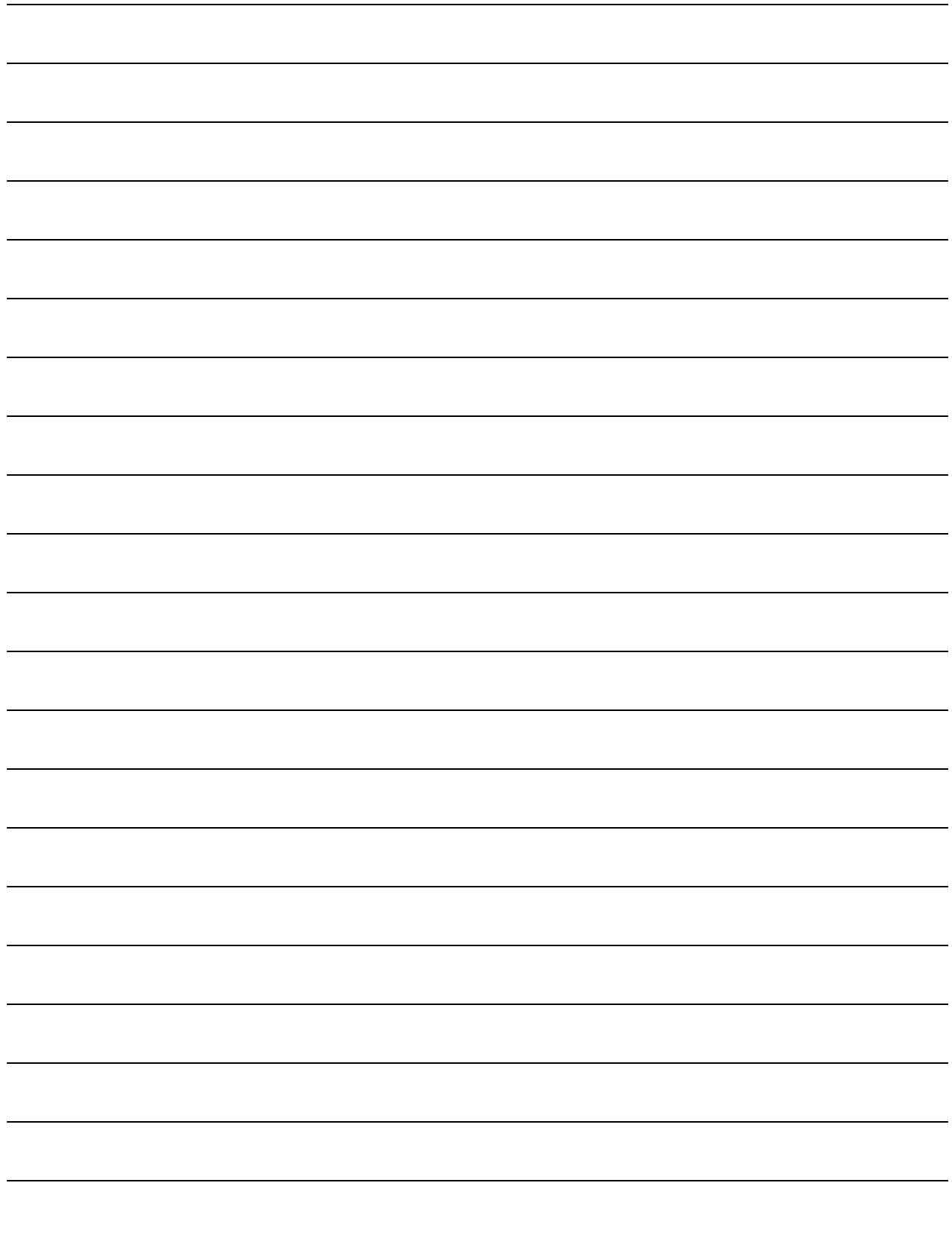
ZOPA: ZOPA stands for Zone Of Possible Agreement. ZOPA is an area where parties are willing to strike a deal.

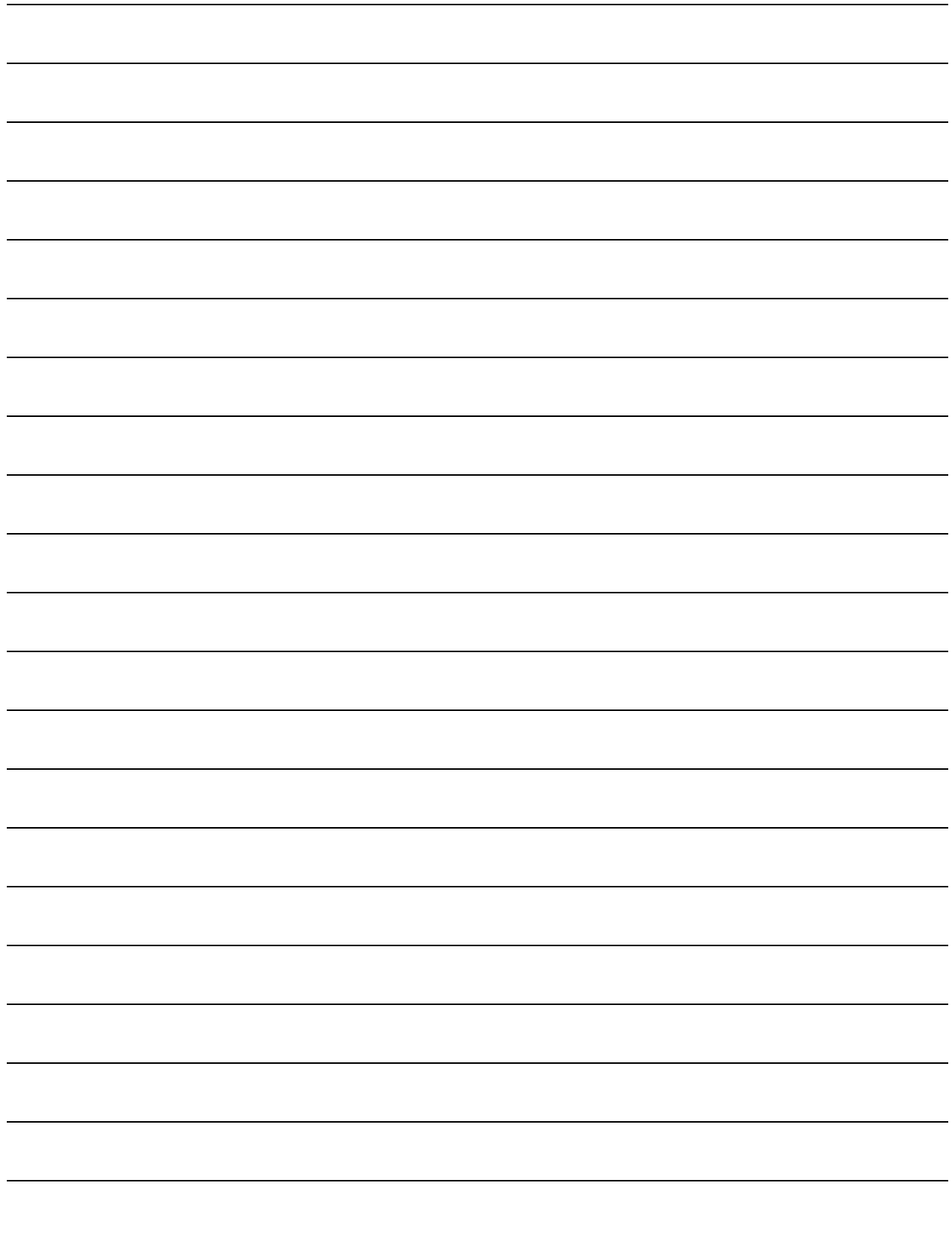
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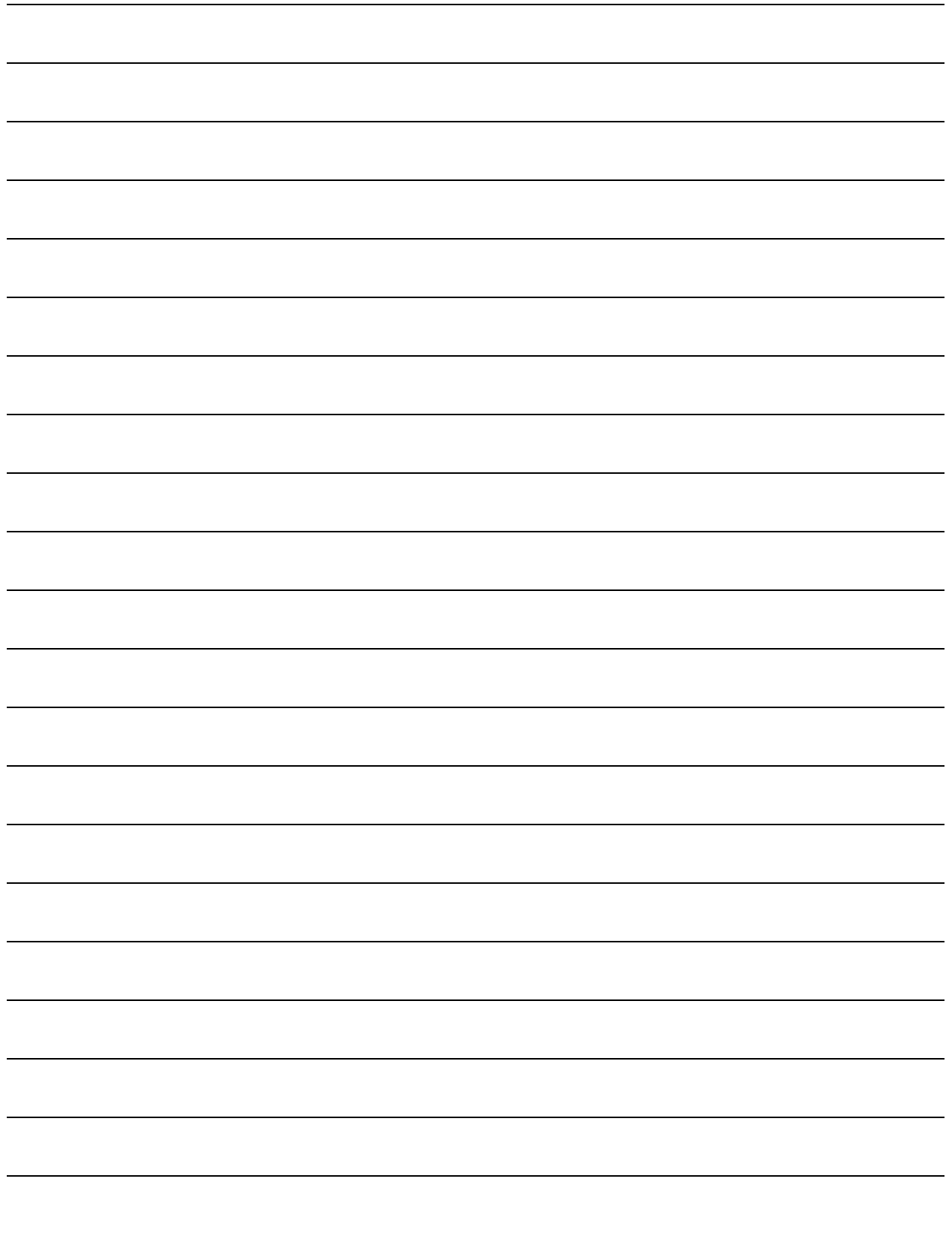
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ABOUT THE AUTHOR



Mittal Shah is an Energy Trading and Risk Management Subject Matter Expert with working experience working in many leading Oil and Gas companies worldwide. His book explains the basics of Energy Trading and Risk Management. Mittal Shah's language is lucid, and the concepts are well explained and can be understood by anyone interested in the topic. Students and IT professionals will benefit tremendously in their careers and this is a must-read to gain domain and contextual knowledge.